

DNA methylation-environment interactions in the human genome

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 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Reviewed Preprint

Revised by authors after peer review.

About eLife's process

Reviewed preprint version 2

January 26, 2024 (this version)

Reviewed preprint version 1

July 5, 2023

Sent for peer review

May 24, 2023

Posted to preprint server

May 19, 2023

Abstract

Previously we showed that a massively parallel reporter assay, mSTARR-seq, could be used to simultaneously test for both enhancer-like activity and DNA methylation-dependent enhancer activity for millions of loci in a single experiment (Lea *et al.*, 2018). Here we apply mSTARR-seq to query nearly the entire human genome, including almost all CpG sites profiled either on the commonly used Illumina Infinium MethylationEPIC array or via reduced representation bisulfite sequencing. We show that fragments containing these sites are enriched for regulatory capacity, and that methylation-dependent regulatory activity is in turn sensitive to the cellular environment. In particular, regulatory responses to interferon alpha (IFNA) stimulation are strongly attenuated by methyl marks, indicating widespread DNA methylation-environment interactions. In agreement, methylation-dependent responses to IFNA identified via mSTARR-seq predict methylation-dependent transcriptional responses to challenge with influenza virus in human macrophages. Our observations support the idea that pre-existing DNA methylation patterns can influence the response to subsequent environmental exposures—one of the tenets of biological embedding. However, we also find that, on average, sites previously associated with early life adversity are not more likely to functionally influence gene regulation than expected by chance.

eLife assessment

This **important** paper uses a genome-wide, massively parallel reporter assay to determine how CpG methylation affects regulatory sequences that control the expression of human genes. The authors provide **compelling** evidence that methylation not only influences baseline activity of regulatory sequences but also the magnitude of acute responses to environmental stimuli. The findings are of broad interest, and the extensive data set will likely become a key resource for the community.

Introduction

DNA methylation is sensitive to the environment, can remain stable across cell divisions, and in some contexts, can alter gene regulation. Consequently, it has received a high level of attention as a potential pathway linking environmental exposures to downstream trait variation, especially when these exposures are separated temporally, as proposed by the “biological embedding” hypothesis (Hertzman, 1999 [↗](#); Hertzman, 2012 [↗](#)). In a canonical example, gestational manipulation of methyl donors in mice is causally implicated in changes in DNA methylation and expression levels of the *agouti* gene, which in turn affects susceptibility to obesity and insulin resistance later in life (Dolinoy *et al.*, 2006 [↗](#)). However, the mechanism underlying the *agouti* case —“metastable” methylation of an inserted intracisternal A particle [IAP] class retrovirus—is now thought to be unusual (Kazachenka *et al.*, 2018 [↗](#)). Thus, it remains unclear whether the many reports of environment-DNA methylation associations are likely to represent links along a causal pathway to phenotype, or are better viewed as passive biomarkers of exposure.

Experimental tests of the causal impact of DNA methylation on gene regulation indicate that both scenarios are possible. *In vitro* studies reveal that transcription factor binding is frequently (but not ubiquitously) sensitive to DNA methylation (O’Malley *et al.*, 2016 [↗](#); Yin *et al.*, 2017 [↗](#)). More recently, epigenomic editing of DNA methylation marks has been shown to alter the activity of transcription factors important in disease and development *in vivo* (Greenberg and Bourc’his, 2019 [↗](#); Yim *et al.*, 2020 [↗](#)), as well as the formation of CTCF-mediated gene loops (Lei *et al.*, 2017 [↗](#)). However, other epigenetic editing studies show that, even within the same regulatory element, changes in DNA methylation have functional consequences for gene expression at only a subset of sites (e.g. Maeder *et al.*, 2013 [↗](#)). Indeed, in the native genome, enhancer activity appears to be insensitive to DNA methylation levels in the majority of cases (Kreibich *et al.*, 2023 [↗](#)). Time series analyses in human dendritic cells have also shown that, in response to a pathogen challenge, nearly all changes in gene expression precede changes in DNA methylation (Pacis *et al.*, 2019 [↗](#)). Such methylation changes may nevertheless be functionally relevant if they influence the speed and magnitude of subsequent challenges (e.g., by marking the accessibility of latent enhancers across cell divisions) (Kamada *et al.*, 2018 [↗](#); Sun and Barreiro, 2020 [↗](#)). This hypothesis may explain, for example, enhanced transcriptional responses to glucocorticoid stimulation in the descendants of hippocampal progenitor cells previously challenged with glucocorticoids (Provençal *et al.*, 2020 [↗](#)).

Together, these findings suggest that environment-associated DNA methylation marks are mixed sets that include functionally silent sites, sites that are constitutively important for gene regulation, and sites in which DNA methylation levels affect gene regulation only under certain conditions. Disentangling which sites belong to which set is important for interpreting and prioritizing the results of a growing body of studies in the biological, social, and health sciences, especially those that test hypotheses that assume a functional role for DNA methylation, such as biological embedding (Hertzman, 2012 [↗](#); Demetriou *et al.*, 2015 [↗](#); Aristizabal *et al.*, 2020 [↗](#)). To facilitate this process, we recently introduced a massively parallel reporter assay, mSTARR-seq, that is capable of testing the functional effects of DNA methylation on regulatory activity in high-throughput (Lea *et al.*, 2018 [↗](#)). This method allowed us to test for methylation-dependent regulatory activity at millions of CpG sites in the human genome. However, our previous data set did not investigate how DNA methylation marks affect the response to environmental challenge. It also did not explicitly target the CpG sites that are the most commonly assayed in the human genome (i.e., those featured on the Illumina MethylationEPIC array).

Here, we address these gaps by using mSTARR-seq to construct a genome-wide map of DNA methylation-dependent enhancer activity for 27.3 million CpG sites in the human genome, accomplished by assessing 600 bp windows encompassing 93.5% of CpGs in the human genome as

a whole. This set includes 99.3% of CpG sites present on the Illumina MethylationEPIC array, the most commonly used platform for profiling DNA methylation in humans, and 99.4% of CpG sites accessible via reduced-representation bisulfite sequencing (RRBS; Meissner *et al.*, 2005 [↗](#)). To evaluate the importance of DNA methylation in responses to environmental challenge, we performed mSTARR-seq under baseline conditions (i.e., no challenge) and following exposure to two physiologically relevant environmental challenges: dosage with the synthetic glucocorticoid dexamethasone, which plays a central role in metabolic regulation and stress-related homeostasis, and interferon alpha (IFNA), a cytokine that elicits genomic and immunological responses associated with viral infection.

We used these data to pursue three goals. First, we describe overall patterns of regulatory activity and methylation-dependent regulatory activity across profiled sites, including how sites targeted by the EPIC array or by RRBS compare to the genome as a whole. The results of this analysis produce a new resource: a map of DNA methylation-dependent enhancer activity across the human genome. Second, we test the degree to which methylation-dependent regulatory activity is affected by exposure to steroid hormone or immune defense-related signaling molecules. Our results yield insight into the frequency of DNA methylation-environment interactions genome-wide, and provide support for the hypothesis that DNA methylation potentiates the cellular response to external stressors. Finally, we illustrate the applicability of this resource by testing two predictions of the biological embedding hypothesis: that pre-existing DNA methylation levels can affect the transcriptional response to subsequent environmental challenge (Provençal *et al.*, 2020 [↗](#); Sun and Barreiro, 2020 [↗](#); Fanucchi *et al.*, 2021 [↗](#)), and that sites linked to early life adversity (ELA) are likely to be functionally important for shaping gene expression. Our results suggest that mSTARR-seq can identify DNA methylation-environment interactions that also occur *in vivo* in human populations and can therefore help prioritize environment- and ELA-associated loci for future follow-up.

Results

mSTARR-seq captures enhancer activity and methylation-dependent enhancer activity genome-wide

We assessed regulatory activity and methylation-dependent regulatory activity for 4,558,475 600-base pair windows of the genome by pairing hybridization capture of targeted loci in the human genome with the massively parallel reporter assay, mSTARR-seq (Lea *et al.*, 2018 [↗](#)). In brief, mSTARR-seq performs enzymatic manipulation of DNA methylation across hundreds of thousands to millions of reporter DNA fragments simultaneously. By measuring the ability of fragments to self-transcribe (as in Arnold *et al.*, 2013 [↗](#); Shlyueva *et al.*, 2014 [↗](#); reviewed in Gallego Romero and Lea, 2022 [↗](#)), it generates estimates of the enhancer-like regulatory potential for each fragment when that fragment is in an unmethylated versus a methylated state (Figure 1A, B [↗](#)). Notably, results from the unmethylated condition are akin to those from a conventional STARR-seq experiment, in that they assess regulatory activity irrespective of CpG content or methylation status. To focus on the CpG sites most likely to be included in DNA methylation studies in humans, we performed custom capture with SeqCap EZ Prime Choice Probes (Roche), targeting all CpG sites on the Illumina Infinium MethylationEPIC array and those likely to be profiled using reduced representation bisulfite sequencing, which enriches for CpG sites near targets of *MspI* restriction enzyme digestion (Figure 1-figure supplement 1 [↗](#)). Because of substantial attention to potential environmental and early life effects on DNA methylation at the glucocorticoid receptor gene (*NR3C1*, reviewed in Liu and Nusslock, 2018 [↗](#)), we also targeted the 6.5 Mb in and flanking *NR3C1*. Finally, to assess background expectations for regulatory and methylation-dependent regulatory activity in the human genome, we targeted a control set of 100,000 loci for capture, chosen at random across the human genome after excluding centromeres, gaps, and uncalled bases in the hg38 genome.

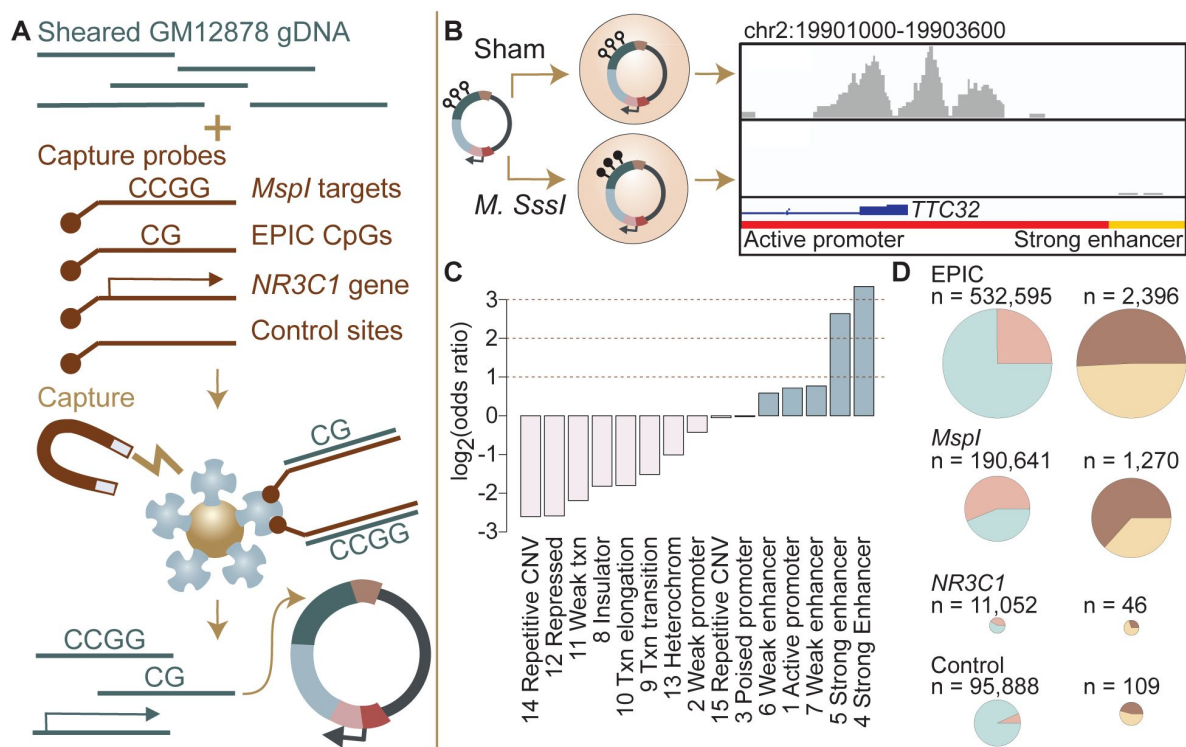


Figure 1.

Study design and identification of enhancer activity.

(A) Sheared DNA from the GM12878 cell line was subjected to enrichment via capture with probes targeting loci selected in reduced representation bisulfite sequencing (RRBS) workflows (*MspI* targets), CpG sites on the Infinium EPIC array, the gene *NR3C1* and flanking regions, and 100,000 randomly distributed control regions. Note that a single 600 bp window can contain multiple target types (Figure 1-figure supplement 1). Library diversity summaries are shown in Figure 1-figure supplement 2. (B) Captured loci were cloned into the mSTARR-seq vector, *pmSTARRseq1*, treated with either the CpG methylating enzyme *M. SssI* or a sham treatment, and transfected into K562 cells. Methylation levels post-transfection and rarefaction analyses of sequencing depth of replicate samples are shown in Figure 1-figure supplements 3, 4. Right panel shows an example of DNA methylation-dependent regulatory activity near the first exon of the *TTC32* gene, where the methylation-dependent regulatory element overlaps an active promoter chromatin state (red horizontal bar denotes active promoter as defined by ENCODE: ENCODE, 2012). (C) mSTARR-seq regulatory activity in the baseline condition is strongly enriched in ENCODE-defined enhancers and some classes of promoters (indicated in blue), and depleted in repressed, repetitive, and heterochromatin states. See Supplementary file 4 for full results of this analysis. Regions with mSTARR-seq regulatory activity detected in this experiment also significantly overlap with regions with regulatory activity in other mSTARR-seq and conventional STARR-seq datasets (Figure 1-figure supplement 5; see also Figure 1-figure supplement 6 for estimates of concordance across technical replicates). (D) Left column shows, under the baseline condition (i.e. unstimulated cells), the proportion of 600 bp windows that exhibited minimal regulatory activity (at least 3 replicate samples produced non-zero RNA-seq reads in either the methylated condition or the unmethylated condition) in the mSTARR-seq assay (pink) versus those with detectable input DNA but no evidence of regulatory activity (blue), for windows containing sites from each target set. Right column shows the proportion of windows with regulatory capacity (i.e., the subset of the windows represented in pink on the left that produce excess RNA relative to the DNA input at FDR < 1%) that are also methylation-dependent (dark brown). Within each column, pie charts are scaled by the total numbers of windows represented. See Figure 1-figure supplements 7 and 8 for comparisons to regulatory regions in other datasets. Figure 1-figure supplements 9 and 10 show window-level RNA to DNA ratios. Figure 1-figure supplement 11 shows the relationship between CpG density and methylation-dependent regulatory activity.

We generated an mSTARR-seq library from captured DNA from the GM12878 cell line, followed by transfection into the K562 erythroleukemic cell line and co-purification of plasmid-derived RNA and DNA (6 replicates per condition: methylated versus unmethylated; see Materials and Methods; Supplementary File 1). Based on our minimal criteria for assessment (600 basepair windows where at least half of the plasmid-derived DNA samples had non-zero reads covering the window; see Materials and Methods), 90.2% of the human genome was included in the plasmid DNA library purified at the end of the mSTARR-seq experiment (mean sequencing depth per replicate = 30.4 million reads; mean read coverage per window = 12.723 ± 41.696 s.d.). Within this set, which also included off-target regions, windows were ~15-fold enriched for targeted regions in the genome (Fisher's Exact Test $\log_2(\text{OR})$ [95% confidence interval (CI)] = 3.971 [3.934, 4.009], $p < 1.0 \times 10^{-300}$). Thus, in windows that passed our minimal assessment criteria, we successfully targeted 99.3% of CpG sites on the MethylationEPIC array and 99.4% of sites likely to be included in RRBS libraries. Across target sets, read depth was not predicted by CpG density, indicating no systematic power differences in assessing methylation-dependent activity due to differences in CpG number per window ($R^2 = 0.099$, $p = 0.6852$; Supplementary file 2). Our results are comparable to published fragment diversity levels achieved with this method (Lea *et al.*, 2018; **Figure 1-figure supplement 2**). Because demethylation or remethylation of DNA fragments could occur within cells during the experiment, we performed bisulfite sequencing at the end of the experiment. We confirmed that DNA methylation levels were substantially higher in the methylated condition samples than in the unmethylated samples, where methylation levels were near zero (mean methylated = 0.885, mean unmethylated = 0.066, **Figure 1-figure supplement 3**; unpaired t-test: $t = -14.66$, $df = 15,124$, $p = 2.39 \times 10^{-10}$). We note that the observed small deviations from 0% and 100% methylation would reduce our power to detect methylation-dependent activity, but should not incur false positive results.

To test for regulatory potential, we focused on those windows where at least 3 replicate samples produced non-zero RNA-seq reads in either the methylated condition or in the unmethylated condition ($n = 216,091$ non-overlapping 600 bp windows, 4.3% of the human genome; mean DNA-seq read coverage per window = 93.907 ± 10.091 s.d.; mean RNA-seq read coverage per window = 165.093 ± 1008.816). These inclusion criteria focused our analysis on the subset of windows where we were able to detect minimal evidence of transcription from the plasmid. The proportion of the human genome showing evidence of regulatory activity in STARR-seq family assays is expected to be small based on previous work. For example, Johnson and colleagues estimated that 0.165% of the human genome had regulatory potential in unstimulated A549 cells in a genome-wide STARR-seq assay, based on the proportion of hg38 basepairs that fell in regions with regulatory activity in their assay (Johnson *et al.*, 2018). Among the regions analyzed here, 3,721 windows (1.7%) showed evidence for regulatory activity, where the amount of RNA generated from a window significantly exceeded the amount of DNA input sequenced for the same window ($\text{FDR} < 1\%$; Supplementary File 3). Sequencing depth does not appear to constrain the ability to detect regulatory regions, as rarefaction analyses suggest that increasing sequencing depth of RNA or DNA libraries would add only a small number of additional regions with any evidence of transcriptional activity (**Figure 1-figure supplement 4**). Further, among analyzed windows, DNA coverage per fragment explains minimal variance in mSTARR-seq enhancer activity ($R^2 = 0.029$, $p < 1.0 \times 10^{-300}$).

Regions with regulatory activity were enriched for enhancers and active promoters annotated by the ENCODE Consortium for the K562 cell line ($\log_2(\text{OR}) = 0.584 - 3.337$; all $p < 1 \times 10^{-3}$; **Figure 1C**; Supplementary file 4 (ENCODE, 2012; Ernst and Kellis, 2012); note that previous research has found that promoters can show enhancer-like activity in STARR-seq-type assays (e.g. Klein *et al.*, 2020). In support of assay reproducibility, regulatory regions also were highly consistent with previous mSTARR-seq data generated in K562 cells using a previously published, independently generated DNA library that we reanalyzed using the current pipeline (overlap

among windows with $\text{FDR} < 1\%$ in each data set analyzed independently: $\log_2(\text{OR})$ [95% CI] = 6.212 [6.086, 6.440], $p < 1.0 \times 10^{-300}$; **Figure 1-figure supplements 5** [5](#)–[6](#); Supplementary Files 5–6) (Lea *et al.*, 2018 [5](#)).

Regions with mSTARR-seq-annotated regulatory activity also exhibited a high degree of consistency between cell types. In a comparison against newly generated mSTARR-seq data from the HepG2 liver cell line, using the same mSTARR-seq library as in the previously published K562 experiments (Lea *et al.*, 2018 [5](#)), regulatory regions detected in this study were enriched among regulatory regions detected in HepG2 cells ($\log_2(\text{OR})$ [95% CI] = 3.534 [3.381, 3.684], $p = 5.21 \times 10^{-307}$; Figure 1-figure supplements 5 and 7; Supplementary File 7). They also overlap with regulatory regions identified in a conventional STARR-seq experiment in A549 lung epithelial cells ($\log_2(\text{OR})$ [95% CI] = 2.451 [2.442, 2.461], $p < 1.0 \times 10^{-300}$) (Johnson *et al.*, 2018 [5](#)). Although we prioritized a blood-derived cell line because of the frequency of environmental epigenetic studies done in blood, these results suggest that our findings can be generalized, to some extent, to other cell and tissue types.

Among the 3,721 windows with regulatory activity in either the methylated or unmethylated condition, 1,768 windows (47.5% of regulatory windows; $\text{FDR} < 1\%$) were differentially active depending on condition, pointing to DNA methylation-dependent regulatory activity. This result is also concordant when we apply the current pipeline to previously published data from K562s and our new HepG2 data, which exhibit correlated effects of DNA methylation on regulatory activity in the windows examined in both cases (K562: $R^2 = 0.286$ for 1250 regulatory windows with $\text{FDR} < 1\%$ in both data sets, $p = 3.19 \times 10^{-93}$; HepG2: $R^2 = 0.277$ for 511 regulatory windows with $\text{FDR} < 1\%$ in both data sets, $p = 8.87 \times 10^{-38}$ Figure 1-figure supplement 8). Among methylation-dependent regulatory regions, the majority of regions (1744 windows, 98.6%) were more active in the unmethylated condition than the methylated condition (**Figure 1-figure supplements 9** [9](#), [10](#) [10](#)). Consistent with previous findings (Lea *et al.*, 2018 [5](#)), regulatory regions with more CpGs are more likely to be repressed by DNA methylation (**Figure 1-figure supplement 11** [11](#); Spearman's $\rho = 0.370$, $p = 9.865 \times 10^{-121}$; $n = 3,721$ regions with mSTARR-seq regulatory activity). Regulatory windows showing higher activity in the methylated condition (24 windows) were enriched for binding motifs of the transcription factors p53, which has been previously reported to have increased binding affinity to methylated DNA relative to unmethylated DNA (Kribelbauer *et al.*, 2017 [5](#)) ($\log_2(\text{OR})$ [95% CI] = 2.352 [0.714, 3.767], $p = 2.91 \times 10^{-3}$; Supplementary File 8), and Tfc211, which has been found to recruit Tet2 to mediate enhancer demethylation (Sardina *et al.*, 2018 [5](#)) ($\log_2(\text{OR})$ [95% CI] = 2.615 [0.580, 4.229], $p = 6.84 \times 10^{-3}$).

Commonly studied CpG sites are enriched for methylation-dependent regulatory activity

We compared methylation-dependent regulatory activity between CpGs targeted on the EPIC array, CpGs typically profiled in RRBS libraries (i.e., near *MspI* cut sites), the 6.5 Mb in or flanking the glucocorticoid receptor gene *NR3C1*, and the control set of genome-representative loci. As expected, windows that contained EPIC CpG sites, sites associated with RRBS, or that were located in or near *NR3C1* were significantly more likely to show at least some degree of transcriptional activity (i.e., show plasmid-derived RNA reads in at least half the sham or methylated replicates) than the genomic background (i.e., 98,967 randomly distributed loci: see Materials and Methods), and thus were more likely to be included in our analysis set ($n = 216,091$ windows) for regulatory activity (EPIC: $\log_2(\text{OR})$ [95% CI] = 2.189 [2.152, 2.227], $p < 1.0 \times 10^{-300}$; RRBS: $\log_2(\text{OR})$ [95% CI] = 4.126 [4.086, 4.163], $p < 1.0 \times 10^{-300}$; *NR3C1*: $\log_2(\text{OR})$ [95% CI] = 3.259 [3.193, 3.325], $p < 1.0 \times 10^{-300}$; **Figure 1D** [5](#)). However, conditional on minimal transcriptional activity, windows containing EPIC array sites were not more likely to exhibit *significant* enhancer-like regulatory activity ($\text{FDR} < 1\%$; $n = 3,721$ windows) than background windows of the genome, and RRBS- and *NR3C1*-associated sites were in fact slightly less likely to do so (EPIC: $\log_2(\text{OR})$ [95% CI] = 0.112 [-0.168, 0.405], $p = 0.47$;

RRBS: $\log_2(\text{OR})$ [95% CI] = -0.493 [-0.779, -0.195], $p = 1.11 \times 10^{-3}$; *NR3C1*: $\log_2(\text{OR})$ [95% CI] = -0.730 [-1.263, -0.217], $p = 3.89 \times 10^{-3}$). This result is likely explained by our inclusion criteria, as only regions with evidence for minimal RNA transcription were retained prior to formal analysis.

In contrast, among regions with detectable regulatory activity, 63.3% of those containing RRBS-associated CpG sites exhibited methylation-dependent regulatory activity, compared to 45.9% of control background loci (RRBS against control: $\log_2(\text{OR})$ [95% CI] = 1.025 [0.430, 1.624], $p = 4.39 \times 10^{-4}$; **Figure 1D**). Note that of 109 regulatory windows in the control set, 108 contain at least 1 CpG, so this difference is not because the control set is CpG-free (mean number CpGs per fragment in control set = 13.291 ± 11.179 s.d.; EPIC = 19.641 ± 14.471 ; RRBS = 20.870 ± 13.883 ; *NR3C1* = 7.405 ± 9.409). Neither windows with EPIC CpGs nor windows in or near *NR3C1* were enriched for methylation-dependent regulatory activity relative to the genomic background (EPIC against control: $\log_2(\text{OR})$ [95% CI] = 0.284 [-0.297, 0.871], $p = 0.33$; *NR3C1* versus control: $\log_2(\text{OR})$ [95% CI] = -0.948 [-2.129, 0.178], $p = 0.11$). Consequently, differential methylation identified through RRBS is more likely to be capable of driving differences in gene regulation, as detectable by mSTARR-seq, than differential methylation elsewhere in the genome. We caution, however, that even among RRBS sites, 98.8% do not occur in regions of the genome with detectable regulatory activity in either the methylated or unmethylated conditions, at least in the K562 cell type, and 35.0% of those that fall in putative regulatory elements exhibit no evidence for methylation-dependent activity.

Environmental perturbation reveals cryptic regulatory elements and cryptic effects of DNA methylation

Enhancer activity can be cell type- or environment-dependent (e.g., Ostuni *et al.*, 2013; Johnson *et al.*, 2018; Chaudhri *et al.*, 2020). DNA methylation-dependent enhancer activity may show similar context-dependence, thus potentially accounting for the large number of sites that fall in functionally silent regions described above. However, this possibility has not been systematically tested. To do so, we next compared regulatory activity between the baseline unchallenged condition and cells challenged with interferon alpha (IFNA) or dexamethasone (dex) (**Figure 2A**; **Figure 2-figure supplements 1-2**; Supplementary file 9). Regions that were identified to have regulatory potential in the baseline condition were highly likely to retain regulatory potential after cells were challenged with IFNA or dex (**Figure 2-figure supplement 3**; IFNA $\log_2(\text{OR})$ [95% CI] = 8.639 [8.499, 8.812], $p < 1.0 \times 10^{-300}$; dex $\log_2(\text{OR})$ [95% CI] = 9.640 [9.483, 9.776], $p < 1.0 \times 10^{-323}$; Supplementary Files 10-11). However, environmental challenges also revealed thousands of putative regulatory regions that were undetectable at baseline but active post-stimulation (**Figure 2B**; 1614 IFNA-specific; 1131 dex-specific). Of 4,632 IFNA regulatory regions (<1% FDR), 44.1% are not detectable at baseline at a 1% FDR threshold in the baseline condition, and even with a relaxed baseline FDR of 10%, 25.1% remain undetectable. Of 4,217 dex regulatory regions (<1% FDR), 40.2% are not detectable at baseline (1% FDR), and 31.5% remain undetectable at a baseline FDR of 10%.

Regulatory windows specific to the IFNA treatment were enriched for 33 transcription factor binding motifs (TFBMs) (Supplementary File 12), with strong enrichments detected for TFBMs involved in innate immune defense in general, and interferon signaling specifically (**Figure 2C**). For example, the most enriched motif was the canonical DNA target of interferon signaling, known as IFN-stimulated response elements (ISRE; $\log_2(\text{OR})$ [95% CI] = 3.158 [2.953, 3.358], Bonferroni corrected $p = 7.31 \times 10^{-133}$), followed by binding motifs for several IFN-regulatory factors (IRF1, IRF2, IRF3, IRF4, IRF8; all $\text{OR} > 1.5$ and Bonferroni corrected $p < 1 \times 10^{-15}$) (Chen *et al.*, 2017). Regulatory windows specific to the dex-stimulated condition were significantly enriched for 28 transcription factor binding motifs, including the glucocorticoid response element IR3 and binding motifs of several transcription factors known to interact with or be modulated by the glucocorticoid receptor (AP-1, CEBP:CEBP, CEBP:AP1, JunB, Jun-AP1, GATA1, STAT3, and STAT5; all $\text{OR} > 1.3$, Bonferroni corrected $p < 0.01$; Supplementary File 13) (Cain and Cidlowski, 2017).

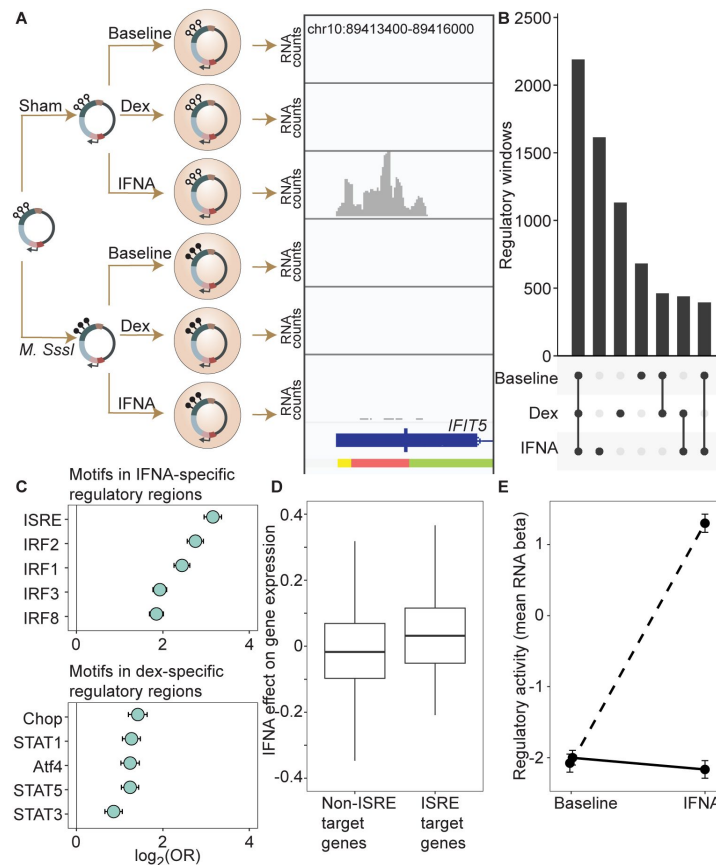


Figure 2.

DNA methylation-environment interactions reveal methylation-dependent responses to IFNA and dexamethasone challenge.

(A) Full mSTARR-seq design across DNA methylation and challenge conditions (see **Figure 2-figure supplements 1** and **2** for filtering and overlap of the datasets, and Supplementary Files 3, 10, and 11 for effect sizes). An example of a DNA methylation-environment interaction is shown overlapping the interferon-induced gene *IFIT5* and an ENCODE-annotated weak promoter (pink denotes weak promoter, yellow denotes heterochromatin, and green denotes weak transcription; the endogenous *IFIT5* gene expression response to IFNA in our experiment is shown in **Figure 2-figure supplement 4**). Three consecutive 600 bp windows have interaction FDR $< 1 \times 10^{-4}$ in this region. Panels depict non-normalized, raw read pileups for mSTARR-seq RNA replicates, with all y-axis maximums set to 14000. No methylation-dependent activity is detectable in the baseline condition because this enhancer element is inactive. Upon IFNA stimulation, only unmethylated enhancer elements are capable of responding. **(B)** Upset plot showing shared and unique mSTARR-seq identified enhancer elements across conditions. While many elements are shared, 3426 are unique to a single condition (FET log-odds results for magnitude of overlap are shown in **Figure 2-figure supplement 3**). **(C)** Top five most enriched transcription factor binding motifs in IFNA- and dex-specific mSTARR-seq enhancers, compared to all windows tested. Whiskers show the 95% CI. See Supplementary Files 12 and 13 for all enrichment results. **(D)** Genes targeted by ISRE enhancers (ISRE enhancers identified from ENCODE ChIP-seq data; gene targets identified from enhancer-gene linkages from EnhancerAtlas 2.0: Gao *et al.*, 2016) that are also identified as IFNA condition-specific mSTARR-seq enhancers show stronger K562 endogenous gene expression responses to IFNA stimulation than non-ISRE targets (unpaired t-test: $t = 3.58$, $df = 118.36$, $p = 5.01 \times 10^{-4}$; Supplementary File 16). Each box represents the interquartile range, with the median value depicted as a horizontal bar. Whiskers extend to the most extreme values within 1.5x of the interquartile range. **(E)** mSTARR-seq regulatory activity for windows containing ISRE targets ($n = 1005$ windows) interacts strongly with exposure to IFNA. These windows are capable of mounting a strong response to IFNA stimulation when unmethylated (dashed line; paired t-test: $t = 23.02$, $df = 1004$, $p = 1.78 \times 10^{-94}$) but not when methylated (solid line; paired t-test: $t = -1.74$, $df = 1004$, $p = 0.082$). Dots show the mean beta corresponding to enrichment of RNA reads versus DNA reads across windows; whiskers show the standard error. Because y-axis values correspond to model estimates, they can be positive (i.e., more mSTARR-seq RNA reads than input DNA reads, indicating no regulatory activity) or negative values (i.e., fewer mSTARR-seq RNA reads than mSTARR-seq input DNA reads, indicating no regulatory activity).

Results were qualitatively similar if we used a more stringent definition of IFNA-specific and dex-specific regulatory activity (e.g. “IFNA-specific” defined as $\text{FDR} < 1\%$ in IFNA condition and $\text{FDR} > 10\%$ in the other two conditions; Supplementary Files 14-15).

To evaluate the relevance of these regions to *in vivo* gene regulation, we also generated matched mRNA-seq data for the endogenous K562 genome from the same experiments. These data showed that expressed genes located closest to mSTARR-seq-annotated, IFNA-specific enhancers were more strongly upregulated after IFNA stimulation than the set of expressed genes located closest to shared mSTARR-seq-annotated enhancers (i.e., those identified in both the IFNA and at least one other condition, considering genes within 100 kb maximum distance from each enhancer element; unpaired t-test; $t = 3.268$, $\text{df} = 601.88$, $p = 1.15 \times 10^{-3}$). We found similar results when using inferred enhancer-gene linkages from the EnhancerAtlas 2.0 (Gao *et al.*, 2016) and restricting the set of IFNA-specific enhancers to those with external experimental support for ISRE binding ($n = 1005$ windows; ChIP-Seq data from ENCODE [2012], relative to all other genes in the gene expression dataset; unpaired t-test; $t = 3.579$, $\text{df} = 118.36$, $p = 5.01 \times 10^{-4}$; **Figure 2D**; Supplementary File 16). Despite a weaker overall gene expression response to dex, genes closest to dex-specific mSTARR-seq enhancers were also more strongly upregulated after dex stimulation than genes near shared enhancers (unpaired t-test; $t = 3.477$, $\text{df} = 479.6$, $p = 5.53 \times 10^{-4}$).

For the IFNA challenge, the DNA methylation state of each window appears to play an important role in shaping condition-specific responses to stimulation in the mSTARR-seq data set. Nearly twice as many regulatory windows (81.4%; 1314 of 1614) exhibited methylation-dependent regulatory activity in the IFNA-specific condition than in the baseline or dex-specific condition (47.5% and 48.4% respectively; two-sided binomial test for IFNA compared to baseline: $p = 3.58 \times 10^{-312}$). Further, regulatory regions that harbor TFBMs for TFs central to the interferon response (ISRE, IRF1, IRF2, IRF3, IRF4, IRF8; $N = 663$ windows) strongly responded to IFNA challenge if in an unmethylated state, but mounted systematically attenuated responses if in a methylated state. As a result, 562 of these 663 windows (84.7%) exhibited significant methylation-dependent regulatory activity and 561 of them (99.8%) were more active in the IFN-challenged state when unmethylated. This pattern is recapitulated when focusing on analyzed windows with external experimental support for ISRE binding ($n = 1005$ windows; ChIP-Seq data from ENCODE [2012]). These windows show no evidence for methylation-dependent regulatory activity in the baseline condition (paired t-test; $t = -0.792$, $\text{df} = 1004$, $p = 0.43$), primarily because they show no regulatory activity at all without IFNA stimulation. After IFNA stimulation, though, they exhibit strong methylation-dependence. Specifically, only unmethylated windows are capable of a response (paired t-test; $t = 31.748$, $\text{df} = 1004$, $p = 1.02 \times 10^{-153}$; **Figure 2E**).

Methylation levels at mSTARR-seq IFNA-specific enhancers predict the transcriptional response to influenza virus in human macrophages

Our findings show that, in the context of mSTARR-seq, pre-existing DNA methylation state can capacitate or constrain the regulatory response to IFNA stimulation. This result suggests that DNA methylation-environment interactions may be an important determinant of gene expression levels *in vivo*. To test this possibility, we drew on matched whole genome bisulfite sequencing (WGBS) and RNA-seq data collected from human monocyte-derived macrophages ($n = 35$ donors), with and without infection with the influenza A virus (IAV), commonly known as flu (**Figure 3A**; Aracena *et al.*, in press). We asked whether variation in the gene expression response to flu is predicted by DNA methylation levels in the baseline (non-infected) condition, especially at loci where the response to IFNA is affected by DNA methylation in mSTARR-seq. Importantly, flu and IFNA challenges induce similar innate immune responses (Killip *et al.*, 2015).

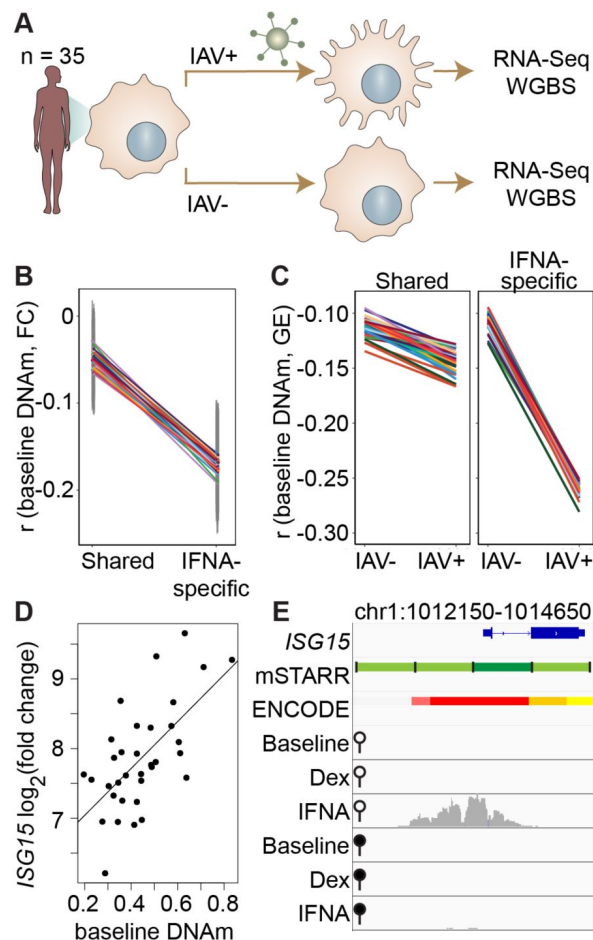


Figure 3.

DNA methylation in mSTARR-seq enhancers predicts *in vivo* gene expression in macrophages.

(A) Study design of the *in vivo* experiment, in which matched macrophage samples from 35 individuals were either left non-infected or infected with influenza A virus (IAV) for 24 hours and processed for RNA-seq and whole genome bisulfite sequencing (WGBS; Aracena *et al.*, in press). **(B)** Within individuals, DNA methylation (DNAm) levels at mSTARR-seq enhancers in non-infected cells are negatively correlated with the nearest genes' transcriptional responses to IAV, but only in mSTARR-seq enhancers that were specific to the IFNA condition (IFNA-specific enhancers: $n = 1033$, mean Pearson's $r = -0.170 \pm 0.009$ s.d., all Bonferroni-corrected $p < 2 \times 10^{-5}$; shared enhancers: $n = 1736$, mean Pearson's $r = -0.049 \pm 0.01$ s.d., all Bonferroni-corrected $p > 0.1$). Each colored line represents an individual, and vertical gray lines represent 95% confidence intervals (see Supplementary File 17 for full results). **(C)** The average within-individual correlation (r) between DNA methylation and gene expression (GE) is 2.44 times as large after infection ($r = -0.261 \pm 0.006$) than at baseline ($r = -0.106 \pm 0.008$) in IFNA-specific mSTARR-seq enhancers (right panel), but much less affected by infection at mSTARR-seq enhancers that are shared across conditions (left panel). Within each panel, each colored line represents an individual for the same set of enhancers (see Supplementary File 18 for full results), without and with IAV infection. **(D)** Across individuals, the *ISG15* transcriptional response to IAV is significantly correlated with average DNAm at the mSTARR-seq enhancer chr1:1013400-1014000 in non-infected cells ($R^2 = 0.381$, $p = 6.05 \times 10^{-5}$, $q = 0.084$). Each dot represents an individual (see Supplementary File 19 for full results and **Figure 3-figure supplement 1** for condition-specific results). **(E)** The mSTARR-seq enhancer predictive of *ISG15* response to IAV (dark green bar) is located in the active promoter of *ISG15* (as defined by ENCODE [ENCODE, 2012]); red denotes active promoter, pink denotes weak promoter, orange denotes strong enhancer, yellow denotes weak enhancer). Three adjacent, methylation-dependent, IFNA-specific mSTARR-seq enhancers were identified (light green), but do not significantly predict *ISG15* response to IAV ($q > 10\%$). The bottom 6 tracks depict non-normalized, raw read pileups for mSTARR-seq RNA replicates in either the unmethylated (open circle) or methylated (filled circle) condition, with all y-axis maximums set to 20,000.

Within each individual in the macrophage data set, mean DNA methylation levels at baseline significantly predict the mean gene expression response to flu across the full set of mSTARR-seq enhancer windows detected in the IFNA condition: higher methylation at baseline predicts an attenuated gene expression response, on average ($n = 35$ individuals at 2769 enhancer windows: mean Pearson's $r = -0.105 \pm 0.006$ s.d., all Bonferroni-corrected $p < 3 \times 10^{-5}$; Supplementary File 17). This effect is largely driven by the subset of mSTARR-seq IFNA-specific enhancers. Specifically, the correlation between baseline DNA methylation levels and the gene expression response to flu is 3.44-fold stronger in IFNA-specific enhancers than for enhancers identified in both the IFNA condition and at least one other condition (**Figure 3B**; IFNA-specific enhancers: $n = 1033$, mean Pearson's $r = -0.170 \pm 0.009$ s.d., all Bonferroni-corrected $p < 2 \times 10^{-5}$; shared enhancers: $n = 1736$, mean Pearson's $r = -0.049 \pm 0.01$ s.d., all Bonferroni-corrected $p > 0.1$). These results appear to be driven by strong methylation-dependence in flu-infected cells, as the average within-individual correlation between DNA methylation and gene expression is 2.44 times as large after infection ($r = -0.261 \pm 0.006$) than at baseline ($r = -0.106 \pm 0.008$) in IFNA-specific mSTARR-seq enhancers (**Figure 3C**; Supplementary File 18).

The limited sample size of the macrophage data set makes it better suited for analyses of the overall relationship between DNA methylation and gene expression within an individual, rather than locus-specific analyses of interindividual variation (especially as locus-specific variance in DNA methylation across individuals is low: mean = 0.004, standard deviation = 0.008). Nevertheless, across 1382 testable loci (600 bp windows containing at least 1 CpG with interindividual variance > 0.01), we identified one IFNA-specific, methylation-dependent mSTARR-seq enhancer where endogenous variation in DNA methylation levels across individuals clearly predicts the response of the nearest gene's transcriptional response to flu (**Figure 3D-E**; $p = 6.05 \times 10^{-5}$, q -value = 0.0837; Supplementary File 19). This mSTARR-seq enhancer (chr1:1013400-1014000) overlaps the promoter of interferon-stimulated gene 15 (*ISG15*; **Figure 3E**), and its average methylation explains 38% of the variance in the *ISG15* transcriptional response to flu across individuals. The magnitude of *ISG15* transcriptional response to flu appears primarily driven by the enhancer's effect on gene expression at baseline. Lower enhancer methylation at baseline is associated with higher *ISG15* expression at baseline, ultimately resulting in a shallower fold change response to flu (**Figure 3-figure supplement 1**). This example illustrates the value of integrating observational data on DNA methylation and gene expression with mSTARR-seq. The population data support the *in vivo* relevance of the mSTARR-seq data, while the mSTARR-seq data suggest that baseline variation in *ISG15* methylation *in vivo* is causally meaningful to the response to influenza. Notably, *ISG15* plays critical roles in regulating the type I interferon response and modulating host immunity to both viral and bacterial infections (reviewed in [Perng and Lenschow, 2018](#)).

Most CpG sites associated with early life adversity do not show regulatory activity in K562s

Finally, because changes in DNA methylation are of particular interest to research on the biological embedding of early life experience ([Hertzman and Boyce, 2010](#)), we tested whether DNA methylation differences associated with early life adversity (ELA) translate to functional effects on gene regulation in the mSTARR-seq data. We first performed a literature search to compile CpG sites that have previously been associated with ELA in humans using the Illumina EPIC array or one of its precursors (Infinium Human Methylation 450K and 27K BeadChips). Our search resulted in a total of 27 studies (Supplementary File 20), which together identified 8,525 unique ELA-associated sites.

For 26 of 27 studies, ELA-associated CpG sites were not more likely to occur within putative regulatory windows (detected in either the methylated condition, unmethylated condition, or both) than background chance (**Figure 4**). This pattern was qualitatively consistent regardless of whether we considered baseline, IFNA-, or dex-challenged samples (Supplementary File 20). The

only exception was for a set of differentially methylated regions in the children of mothers exposed to objective hardship (e.g., living in a shelter, loss of electricity) who were pregnant during or within 3 months of the 1998 Quebec Ice Storm (Cao-Lei *et al.*, 2014 [DOI](#)). In this study, ELA-associated sites were more likely to fall in windows with regulatory potential in our sample ($\log_2(\text{OR})$ [95% CI] = 1.343 [0.715, 1.909], $p = 3.39 \times 10^{-5}$). Among these sites, 53.85% were also detected to have methylation-dependent activity, which is slightly, but not significantly higher than the proportion of methylation-dependent sites on the Illumina Methylation450K chip as a whole ($\log_2(\text{OR})$ [95% CI] = 0.884 [-0.338, 2.130], $p = 0.16$). Consequently, ELA-associated sites in this study were not more likely to exhibit methylation-dependent activity than chance. We speculate that regulatory enrichment in this data set is due to its focus on intermediately methylated CpG sites with substantial interindividual variance in DNA methylation levels, which tends to enrich for enhancer elements. Indeed, ELA-associated sites in this study were more strongly enriched in enhancer regions than the union set of sites in other studies we investigated ($\log_2(\text{OR})$ [95% CI] = 1.295 [0.543, 2.015], $p = 6.64 \times 10^{-4}$).

Discussion

Using mSTARR-seq, we assessed the functional role of DNA methylation across more than 99% of CpG sites assessed by two commonly used methods to measure methylation (the EPIC array and RRBS). Compared to our previous work, we identify a higher rate of methylation-dependence in the present data set (47.5% of analyzed windows in the current data set versus 10% of analyzed windows from Lea *et al.* (2018) [DOI](#)) when we apply the current pipeline to analyze both datasets; Supplementary Files 3, 6). This difference likely stems from lower variance between replicates in the present study, which increases power (Figure 1-figure supplement 6 [DOI](#)). Thus, our findings reveal that DNA methylation often significantly attenuates regulatory activity in K562 enhancer and promoter elements.

The results also support the idea that CpG sites identified by environment-DNA methylation association are a mixed bag. For example, despite the pervasive assumption in the literature and in popular science that early adversity causally impacts downstream outcomes through persistent epigenetic modification (e.g., Dubois *et al.*, 2019 [DOI](#)), 98% of CpG sites associated with early life adversity in the literature fell in windows with no discernable regulatory potential in K562 cells, irrespective of their methylation status or cellular state. Our findings suggest that many ELA-associated sites may be better treated as passive biomarkers of exposure rather than links on the causal pathway between early life disadvantage and later life health outcomes. However, other cell types and cellular contexts should be tested to further evaluate this hypothesis (we used the erythroleukemic K562 cell line, as ELA-associated sites are most commonly assessed in blood). For instance, while we included the dexamethasone condition here because glucocorticoid dysregulation is commonly invoked in association with the response to early life adversity, the relationship between glucocorticoid signaling and early life adversity is complex (Eisenberger and Cole, 2012 [DOI](#); Koss and Gunnar, 2018 [DOI](#)) and may not be well-modeled by acute glucocorticoid exposure. Given that the repertoire of poised and active enhancers often differs across cell types and cellular states, expanding experimental approaches like mSTARR-seq to other contexts can therefore serve a valuable role in prioritizing CpG sites identified in observational studies of differential methylation (e.g., CpG sites associated with disease).

Our results dovetail with work supporting the importance of CpG sites within enhancer elements in responding to environmental perturbations (Pacis *et al.*, 2015 [DOI](#); Husquin *et al.*, 2018 [DOI](#)). Indeed, one of the most interesting findings from our analysis is the observation that DNA methylation-environment interactions are widespread in the human genome. Across thousands of genomic regions, both regulatory activity and methylation-dependent effects on regulatory activity were only detectable upon stimulation, consistent with a model in which DNA methylation contributes to epigenomic priming by modulating responsiveness to environmental stimuli (Kamada *et al.*,

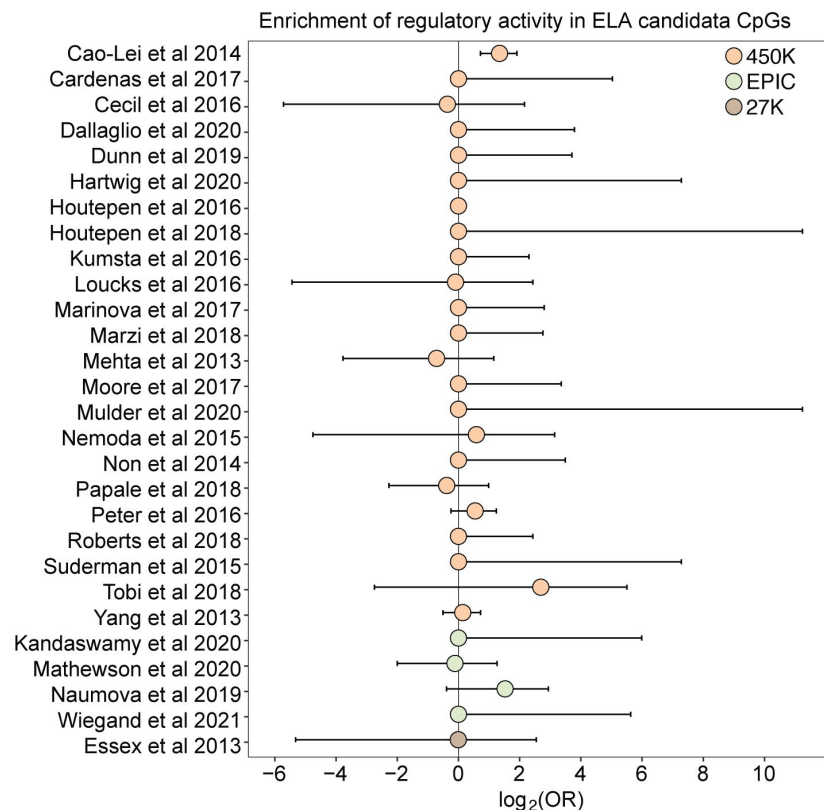


Figure 4.

Early life adversity-associated CpG sites are not enriched for mSTARR-seq regulatory activity.

Log₂-transformed odds ratios from Fisher's Exact Tests for enrichment relative to the background set of sites on each array platform, for 27 studies of early life adversity-DNA methylation level correlations (see Supplementary File 20 for full FET results). Whiskers show standard error. Only Cao-Lei *et al.*, 2014 [\[1\]](#) shows significant enrichment for regulatory activity (log₂(OR) [95% CI] = 1.343 [0.715, 1.909], $p = 3.39 \times 10^{-5}$), but these sites are not more likely to exhibit methylation-dependent activity than expected by chance (log₂(OR) [95% CI] = 0.884 [-0.338, 2.130], $p = 0.16$). For details on the source tissue and measures of ELA, see **Figure 4-figure supplement 1** [\[2\]](#).

2018 [↗](#); Sun and Barreiro, 2020 [↗](#)). In support of this idea, IFNA-specific enhancers detected in mSTARR-seq are able to nonrandomly identify collections of loci in the genome where baseline DNA methylation also predicts the endogenous gene expression response to flu infection (**Figure 3B-C** [↗](#)). This relationship is easier to discern in within-individual comparisons across loci than in locus-by-locus analyses of interindividual variation. However, we were also able to pinpoint one specific gene, *ISG15*, in which baseline DNA methylation explains a large fraction of population variation in the gene expression response to flu (**Figure 3D-E** [↗](#)). Notably, the effect size for *ISG15* is very large, accounting for over a third of total variance in the fold-change gene expression response to flu. Unfortunately, a comparable *in vivo* dataset for dex challenge was unavailable to conduct parallel validation studies. However, our results overall suggest that mSTARR-seq could help identify additional *in vivo* DNA methylation-environment interactions, given sufficiently large sample sizes and data on the environmental exposures of interest.

Applying mSTARR-seq in additional cell types may therefore help resolve whether DNA methylation is responsible for exaggerated transcriptional responses to repeated challenges, as previously suggested for glucocorticoid exposure in hippocampal progenitor cells (Provençal *et al.*, 2020 [↗](#)), or whether differential methylation after pathogen infection contributes to the emerging paradigm of “trained immunity” in innate immune cells (reviewed in Fanucchi *et al.*, 2021 [↗](#)). More generally, while our findings agree with the idea that many differences in DNA methylation—even extreme ones (~0% versus ~100%) like those tested here—are silent with respect to transcription factor binding and gene expression (e.g., Maeder *et al.*, 2013 [↗](#); Kreibich *et al.*, 2023 [↗](#)), they also suggest that the functional importance of DNA methylation is likely to be underestimated without considering its interaction with the cellular environment.

Materials and Methods

DNA capture and mSTARR-seq

We used the DNEasy Blood and Tissue Kit (Qiagen) to extract 5 ug DNA from the GM12878 lymphoblastoid cell line. We sheared the extracted DNA on a Covaris S2 with the following parameters: intensity=3; duty cycle=5%, cycles/burst=200, treatment time=40 seconds, temperature=4° C; intensifier=yes. We then performed agarose gel size selection of ~600-700 bp DNA fragments followed by purification with the Qiaquick Gel Extraction Kit (Qiagen). We note that we intentionally targeted longer fragments (~600-700 bp) than those targeted in (Lea *et al.*, 2018 [↗](#)) (~300-700 bp) because our previous work showed that longer fragments are more likely to drive regulatory activity (Lea *et al.*, 2018 [↗](#)).

We used Roche’s NimbleDesign Software to design probes to capture the following genomic regions: (i) EPIC CpGs targeted by the Illumina Infinium MethylationEPIC microarray (862,831 CpGs); (ii) predicted CpG cut sites from an *in silico* *Msp1* digest of the hg38 genome, followed by *in silico* size selection to 100-500 bp, which together simulates the first step in reduced representation bisulfite sequencing protocols (318,929 CpGs); (iii) 6.5 Mb centered around the *NR3C1* glucocorticoid receptor gene; and (iv) 100,099 loci included as a control for genomic background that were randomly distributed across the hg38 human genome, excluding centromeres, gaps, and Ns. Although the Y chromosome was included during the design of control loci, we excluded the Y chromosome in analyses, resulting in 99,999 control loci which fell into 98,967 unique 600 bp windows. Of these 98,967 windows, 94,679 contained CpG sites and 4,288 did not, reflecting the sequence composition of the human genome as a whole. We note that a single 600 bp genomic window could simultaneously include EPIC CpGs, *Msp1* CpG cut sites, the *NR3C1* region, or control sites (i.e., assignment of windows to these compartments are not mutually exclusive; **Figure 1-figure supplement 1** [↗](#)).

We then captured target regions from the sheared GM12878 DNA using SeqCap EZ Prime Choice Probes (Roche), following the SeqCap EZ HyperCap protocol version 2.3. We performed two capture reactions. We used 6 cycles for the Pre-Capture LM-PCR program and 16 cycles for the Post-Capture LM-PCR program. For the Post-Capture LM-PCR reaction, we used the mSTARR_primerF2 and mstarr_primerR2 primers from [Lea et al. \(2018\)](#) to prepare the DNA for ligation into the mSTARR plasmid. The resulting captured DNA was used as input for the mSTARR-seq protocol ([Lea et al., 2018](#)), beginning at the mSTARR-seq published protocol step “Generate linearized mSTARR backbone for large-scale cloning”.

We performed 4 and 6 replicate Gibson assemblies and transformations for the first and second capture reactions, respectively. We then sequenced each transformation replicate on a MiSeq to measure library diversity per replicate, with the following modifications to the published mSTARR-seq protocol: we used 25 ul Kapa HiFi HotStart ReadyMix instead of 25 ul NEB Q5 Hot start master mix, with the following cycle program: 98 C for 45 sec, 12 cycles of 98 C for 15 seconds, 60 C for 30 seconds, and 72 C for 60 seconds, followed by a final extension at 72 C for 5 minutes. We measured library diversity of each transformation using the ENCODE Project’s PCR bottleneck coefficient (PBC), which is the percentage of non-duplicate mapped reads from the total number of mapped reads ([Landt et al., 2012](#)) (Supplementary File 21). We pooled the ten transformations by weighting their molarities according to the PBCs to create the final library. We then performed 8 re-transformations to expand the pooled library.

Replicate methyltransferase reactions and parallel mock methyltransferase reactions, in which the *M. SssI* enzyme was replaced with water, were performed in 500 ul reactions, cleaned with Ampure beads, and then pooled. The methylated DNA library and unmethylated DNA library were each transfected into 18 replicate T75 flasks, each containing 12 million K562s, with Lipofectamine 3000 (ThermoFisher Scientific) following the manufacturer’s instructions. 42 hours post-transfection, replicates were treated with 2,000 U/mL IFNA2b (Thermo Fisher Scientific), 1 uM dex (Sigma-Aldrich), or vehicle control (media). 48 hours post-transfection (6 hours post-treatment), cells were harvested for mSTARR-seq sequencing library generation. At harvest, 5×10^5 cells were aliquoted separately into Buffer RLT to measure endogenous RNA response to treatment, and 2 million cells were aliquoted for plasmid DNA extraction to measure input DNA in each replicate. mSTARR RNA-seq and DNA-seq libraries were generated following [Lea et al. \(2018\)](#). To measure endogenous gene expression from K562s in each condition, we extracted RNA from the separately aliquoted cells using the Qiagen RNEasy kit and prepared RNA-seq libraries using the NEBNext Ultra II RNA Library Prep Kit for Illumina.

Comparison of mSTARR MD regulatory activity across cell types and experiments

To assess cell type-specificity of methylation-dependent regulatory activity, we transfected the mSTARR-seq library from [Lea et al. \(2018\)](#), comprised of 1:3 sheared genomic DNA:*MspI*-digested DNA, into the HepG2 cell line. For the HepG2 experiment, we performed one methyltransferase reaction and one sham methyltransferase reaction. Transfections were performed with Lipofectamine 3000 (ThermoFisher Scientific) following the manufacturer’s instructions, with reagent quantities scaled to the following per replicate: 6.9 million HepG2 cells, 46 ug of DNA, 138 ul of Lipofectamine 3000, and 180 ul of P3000 (transfection enhancer reagent). We performed all regulatory and MD regulatory analyses of HepG2 cells, as well as analyses of K562s from a previously published smaller mSTARR-seq experiment ([Lea et al., 2018](#)), following the same bioinformatics pipeline used for our main K562 experiment, described below. To compare against previously published STARR-seq data in A549 cells, we identified the union set of all regulatory peaks identified across six replicate STARR-seq experiments collected at time point 0 in [Johnson et al. \(2018\)](#). To test for significant overlap between the current K562 results and the other three data sets, we used Fisher’s exact tests against a background set of windows tested in both the current K562 experiment and each of the other data sets.

mSTARR-seq regulatory and MD regulatory analyses

mSTARR RNA-seq and DNA-seq libraries were sequenced on the Illumina NovaSeq platform as 100 basepair, paired-end reads (Supplementary File 1). Reads were trimmed with Trim Galore (version 0.6.4) (Krueger, 2015) to remove basepairs at the ends of reads with Phred scores less than 20 and stretches of 2 or more basepairs that matched adapter sequences. Trimmed reads with a minimum length of 25 basepairs were retained. We mapped reads using *bwa* (version 0.7.12) (Li and Durbin, 2009) using default settings. We retained read pairs that mapped to a single best location using the *samtools* (version 1.3.1) (Li *et al.*, 2009) package ‘view’ command with options -q 10 -f 0x2. We segmented the genome into 600 bp, non-overlapping windows, and used the *bedtools* (version 2.25.0) (Quinlan and Hall, 2010) *coverage* function to count the number of RNA and DNA fragments overlapping each 600 bp window. We chose 600 bp for the genomic window size to accommodate our library’s DNA fragment size (600-700 bp; Supplementary File 1), and because using smaller windows resulted in the identification of many adjacent windows as separate regulatory elements when they likely represent a single true enhancer (the median length of ENCODE-annotated enhancers is 600 bp).

For each of the 3 treatment conditions (baseline, IFNA, and dex), we reduced our dataset to windows for which at least three DNA samples in the methylated condition and three DNA samples in the unmethylated condition (i.e., 6 DNA samples total), and three RNA samples in *either* the methylated or unmethylated condition, had nonzero counts. One mSTARR RNA-seq sample in the baseline condition (sample ID L31250) was removed from further analysis because it had an unusually high proportion of zero counts in the testable windows; the corresponding paired DNA sample was therefore also removed prior to analysis (Supplementary File 1). Next, we retained only windows that showed high repeatability across DNA samples, following Lea *et al.* (2018). In brief, for each window we calculated the pairwise difference of read abundance for every pair of samples and created a distribution of all pairwise differences; windows were removed if at least 25% of pairs fell outside the central 90th percentile of the distribution (37,410 windows, 14.8% of the overall data set). On this reduced data set of testable windows, we performed voom normalization using the *limma* (version 3.44.3) *voomWithQualityWeights* function (Smyth, 2005; Law *et al.*, 2014), with methylation status included as a covariate in the design. We then used the *limma* function *lmFit* to apply the following model for each window:

$$y_i = \mu + m_i\beta_1 + s_i\beta_2 * I(m = 0) + s_i\beta_3 * I(m = 1) + \varepsilon_i$$

where y_i is the vector of normalized counts for $n = 24$ samples (12 RNA and 12 DNA samples; 22 samples for the baseline condition); μ is the intercept; m_i is methylation condition (0 = unmethylated; 1 = methylated) and β_1 its effect size; and I is an indicator variable in which β_2 and β_3 are the effects of sample type (s_i ; 0 = DNA; 1 = RNA) in the unmethylated ($m = 0$) and methylated conditions ($m = 1$), respectively. ε_i is the residual error.

To estimate the false discovery rate for identifying windows with regulatory activity (in either the methylated or unmethylated condition, or both), we compared the observed results to empirical null distributions generated using 100 permutations of RNA/DNA labels within each pair of samples following Storey and Tibshirani (2003). Unlike in the real data, in which most windows show a strong bias towards more reads in the DNA condition (because most windows exhibit no regulatory activity, and therefore show many more input DNA reads than RNA reads), permuted data results in relatively balanced effect sizes. Using the overall distribution of p-values to construct the null therefore substantially inflates the number of significant windows, but specifically because most observed windows have systematically *less* activity in the RNA condition than the DNA condition, not more (the direction of interest). Thus, retaining all p-values to construct the null leads to highly miscalibrated false discovery rates because the distribution of observed values is skewed towards smaller values—because of windows with “significantly” no regulatory activity—compared to the permuted data. To address this problem, for each permutation, we subsampled N windows with a positive sample type beta (corresponding to larger

read counts in RNA samples versus DNA samples) in either the methylated condition, unmethylated condition, or both. Here, N is the number of windows with a positive sample type beta in the observed data (e.g., $N = 17,461$ in the baseline dataset). The p-values from these windows were used to generate the empirical null. To define putative enhancer elements, we used an FDR cutoff of 1%.

To identify methylation-dependent enhancer activity, we focused on the windows that exhibited a positive sample type beta (i.e., exhibited more RNA reads than DNA reads) in either the methylated or unmethylated condition (or both) and tested for significant differences between β_2 and β_3 in the equation above (i.e., an interaction effect between RNA/DNA condition and the methylated/unmethylated condition). To estimate the false discovery rate for identifying methylation-dependent windows, we re-ran our main nested model on the same 100 sets of N windows each, in this case permuting methylation condition (unmethylated versus methylated) across sample pairs to generate the empirical null for the interaction effect of sample type (RNA versus DNA) * methylation condition (methylated or unmethylated). To define methylation-dependent regulatory elements, we again used an FDR cutoff of 1%.

To assess whether mSTARR RNA-seq and DNA-seq libraries were sequenced deeply enough to saturate detection of unique 600 bp windows included in the formal analyses, we used seqtk (<https://github.com/lh3/seqtk>) to randomly subset sequence reads from the raw fastq files for all RNA-seq or all DNA-seq replicates in the baseline dataset. We then applied to these data subsets the same filtering pipeline described above to generate a reduced data set of testable windows. The results of this rarefaction analysis (shown in **Figure 1-figure supplement 4**) shows that our sequencing depth is sufficient to capture all analyzable windows based on both the DNA read filter and the RNA read filter.

Enrichment of transcription factor binding motifs

To identify enrichment of potential transcription factor binding sites in the methylated condition, we used HOMER and motifs defined in the HOMER database (Heinz *et al.*, 2010) and the baseline treatment dataset to compare motifs within regulatory regions in the methylated condition, relative to motifs across all regulatory regions.

To identify potential transcription factor binding sites in windows with condition-specific regulatory or methylation-dependent regulatory activity, we first identified regions showing regulatory activity uniquely in the dex or IFNA conditions relative to both other conditions (based on an FDR of 1% to define regulatory activity). We used HOMER and motifs defined in the HOMER database to test for enrichment of transcription factor binding motifs within windows showing condition-specific regulatory activity relative to all windows tested for regulatory activity in that condition. We set a threshold of Bonferroni-corrected p-value < 0.01 to identify significantly enriched binding motifs.

Endogenous gene expression response of K562s to dex and IFNA

Endogenous RNA-seq libraries from K562 cells challenged with dex or IFNA were sequenced on an Illumina NovaSeq 6000 S1 flow cell as 100 base pair, paired-end reads. Reads were trimmed with Trim Galore (Trim Galore version 0.6.4_dev; Cutadapt version 2.3) to remove basepairs with a Phred score less than 20, and end sequences that matched at least two basepairs of the adapter sequence. Only trimmed reads longer than 25 basepairs were retained. We used the STAR package (version 2.5.0) (Dobin *et al.*, 2013) two-pass mapping to map the filtered reads to the hg38 genome. We retained uniquely mapped reads by filtering the output SAM file to keep reads with MAPQ = 255. We then used *htseq* (version 0.6.0) (Anders *et al.*, 2014) to quantify read counts per gene. We only retained genes that had TPM > 3 in at least 3 of 6 samples in at least one of the three conditions (baseline, IFNA, or dex). Only protein-coding genes were retained for final analysis, resulting in a total of 10,676 testable genes.

We performed differential expression analysis separately for IFNA and dex, by subsetting the data to the IFNA and baseline samples, or subsetting the data to the dex and baseline samples. For each subset, we *voom* normalized the data (Law *et al.*, 2014 [DOI](#)), and used the *limma* (Smyth, 2005 [DOI](#)) function *lmFit* to model the normalized gene expression as a function of treatment, controlling for the percent of reads mapping to annotated features (to account for technical variation in the efficiency of RNA purification during library prep) and methylation condition for the associated sample. We calculated the FDR using the q-value approach of (Storey and Tibshirani, 2003 [DOI](#)), based on an empirical null distribution that was derived from 100 permutations of treatment condition in the model.

To test whether genes were more responsive to IFNA stimulation if they were predicted targets of IFNA-specific ISRE enhancers, we first used the *bedtools intersect* function to identify the union set of ChIP-seq peaks for STAT1 and STAT2 (two of the three components of the ISGF3 transcription factor that binds ISRE motifs; ENCODE accession numbers ENCFF478XGE and ENCFF394KTR; ChIP-Seq data were not available for the third ISGF3 component, IRF9). We then reduced these regions to those showing significant, IFNA condition-specific regulatory activity in the mSTARR-seq dataset (FDR < 1%). We used K562 enhancer-gene links from EnhancerAtlas 2.0 (Gao *et al.*, 2016 [DOI](#)) to link the resulting IFNA-specific ISRE enhancers to their target genes. Finally, we performed a two-sided, unpaired t-test to compare the endogenous expression responses to IFNA for genes associated with putative IFNA-specific ISRE enhancers relative to genes that are not associated with IFNA-specific ISRE enhancers.

Endogenous gene expression and methylation in human macrophages

To assess effects of DNA methylation-environment interactions on gene expression *in vivo*, we evaluated endogenous methylation and gene expression from matched whole genome bisulfite sequencing (WGBS) and RNA-seq data collected from human monocyte-derived macrophages (n = 35 donors), with and without infection with the influenza A virus (IAV; Aracena *et al.*, in press). Unsmoothed methylation counts were obtained for 19,492,906 loci in both non-infected and IAV-infected samples (total n = 70) as described (Aracena *et al.*, in press). We filtered loci to require coverage of ≥ 4 sequence reads in at least half of the non-infected or IAV-infected samples. In the RNA-seq dataset for the same 35 individuals, we excluded any genes that did not have an average RPKM > 2 in non-infected or IAV-infected samples. This resulted in a total of 19,041,420 CpG sites and 14,122 genes used in downstream analyses.

We calculated normalization factors using *calcNormFactors* in *edgeR* (v 3.28.1) (Robinson *et al.*, 2010 [DOI](#)) to scale the raw library sizes. We then used the *voom* function in *limma* (v 3.42.2) (Smyth, 2005 [DOI](#); Ritchie *et al.*, 2015 [DOI](#)) to apply the normalization factors, estimate the mean-variance relationship, and convert raw read counts to log(CPM) values. Sequencing batches were regressed out using *ComBat* from the *sva* Bioconductor package (v 3.34.0) (Leek *et al.*, 2012 [DOI](#)), which fit a model that also included age (mean-centered) and admixture estimates. We subsequently regressed out age effects using *limma*. Individual-wise fold-change (FC) matrices were built by subtracting non-infected counts from IAV-infected counts for each individual using weights calculated as in (Harrison *et al.*, 2019 [DOI](#); Aracena *et al.*, in press).

For comparison of the mSTARR-seq dataset to the dataset from Aracena *et al.* (in press), GrCh38/hg38 coordinates were lifted over to GRCh37/hg19 using the UCSC liftOver tool. We required a 0.95 minimum ratio of bases that remap, excluded loci that output multiple regions, set the minimum hit size in query to 0, and set the minimum chain size in the target to 0. The GRCh37/hg19 coordinates were used in the following analyses.

We first sought to assess, within each of the 35 individuals, the correlation between mSTARR-seq enhancer methylation levels at baseline (i.e., in non-infected cells) and transcriptional responses of their nearest genes to IAV. To find overlaps between enhancer regions and CpG loci, we used the *bedtools* intersect function (v2.29.1) with the “left outer join” option. For each 600 bp mSTARR-seq enhancer region in each individual, we calculated the mean methylation level across all overlapping CpGs. Each enhancer was linked to its nearest gene, with a maximum distance of 100 kb, as described above. If an enhancer had multiple linked genes (e.g., an enhancer that overlapped more than one gene), we took the mean transcriptional response of all linked genes. For each individual, we calculated the Pearson’s correlation coefficient (r) between the DNA methylation levels within the mSTARR-seq-defined enhancer windows, in non-infected cells, and transcriptional responses of their linked genes to IAV infection. We also investigated the correlation between mean mSTARR-seq enhancer window methylation and gene expression *within* non-infected and flu-infected samples separately.

Finally, for each mSTARR-seq enhancer-gene pair, we sought to test the extent to which average methylation level in the enhancer in non-infected samples explained the gene’s transcriptional response to IAV, across individuals. Here, we calculated the average CpG methylation level for each 600 bp enhancer, after excluding CpGs with methylation variance less than 0.01 across individuals. Thus, enhancers that did not contain any CpGs with appreciable interindividual variation in DNA methylation levels were excluded from the analysis. This filtering step resulted in 1,382 enhancer-gene pairs for this analysis. For each enhancer-gene pair, we calculated the Pearson’s correlation coefficient (and R^2) between the average methylation level of the enhancer and the transcriptional response of the linked gene. P-values were corrected for multiple hypothesis testing using the q-value method in R (Storey and Tibshirani, 2003 [↗](#)).

CpG methylation associated with early life adversity

We performed a literature search to identify studies for which CpG methylation differences have been linked to adverse conditions during early life. We considered all journal articles that contained the words “early life adversity” and “Infinium” on <https://scholar.google.com> [↗](#) on May 10, 2021, which produced 269 results. We required that the study evaluated CpG methylation using an Illumina Infinium array (27K, 450K, or EPIC), and that the article provided Infinium CpG probe IDs or CpG genomic coordinates of candidate CpGs. Articles that reported no significant CpG sites, but still reported “top” CpG sites, were retained. Articles that performed analyses to identify differentially methylated regions (DMRs; as opposed to CpG site-by-site analysis), and then reported individual CpG sites within the candidate DMRs, were retained. We considered early life adversity to encompass both social and nonsocial sources of environmental adversity (e.g., exposure to severe weather), any time from prenatal development to 18 years of age. We did not impose criteria for cell type or subject age at time of sampling. See **Figure 4-figure supplement 1** [↗](#) and Supplementary File 20 for a summary of the resulting studies.

For each study dataset, we performed a Fisher’s exact test to test whether ELA candidate sites, relative to the total set of CpG sites on the Illumina array used in that study (450K, EPIC, or 27K array), were enriched in the mSTARR-seq regulatory windows we identified in the baseline, dex, or IFNA condition.

Acknowledgements

We thank B.J. Nielsen, Tawni Voyles, and Tina Del Carpio for their contributions to data generation, Terrie Moffitt and the Reddy lab at Duke University for project feedback, Graham Johnson for sharing the A549 data, Courtney Karner for use of equipment, Xiang Zhou for constructive advice on statistical modeling, and two anonymous reviewers for constructive feedback on an earlier version of this manuscript. This study was supported by the National Institutes of Health (R01HD088558 to J.T. and F32HD095616 to R.A.J.), the Canadian Institutes for

Advanced Research Child Brain and Development Program, a Sloan Foundation Early Career Research Fellowship to J.T., and a Foerster-Bernstein Postdoctoral Fellowship to R.A.J. High-performance computing resources were supported by the North Carolina Biotechnology Center (Grant Number 2016-IDG-1013 and 2020-IIG-2109).

Data Availability

mSTARR-seq RNA and DNA sequencing data generated in this study for K562 and HepG2 cells are available in the NCBI Sequence Read Archive (SRA; BioProject accession number PRJNA922490). K562 endogenous RNA-seq data are available in the NCBI Gene Expression Omnibus (GEO series accession GSE222643). The previously published data from (Lea *et al.*, 2018 [↗](#)) used in this study are available in NCBI's SRA accession number SRP120556. The previously published A549 data from (Johnson *et al.*, 2018 [↗](#)) were kindly provided by Graham Johnson. The sequencing data for the macrophage RNA-seq and WGBS datasets are available at the European Genome-Phenome Archive (EGA) under accession numbers EGAD00001008422 and EGAD00001008359, respectively. The zenodo repository at <https://zenodo.org/record/7949036#.ZGZ5UnbMjq9> [↗](#) provides the code used for analyses, as well as track files (compatible with the UCSC genome browser) for mSTARR-seq hg38 windows with regulatory activity, and windows with methylation-dependent regulatory activity, in all three conditions (baseline, IFNA, and dex). The mSTARR-seq plasmid, *pmSTARRseq1* is available through AddGene.

Supplementary Files

Supplementary File 1. Table summarizing the K562 mSTARR-seq DNA and RNA libraries generated in this study.

Supplementary File 2. Table providing sequencing depth and CpG composition statistics for the target regions (EPIC array, RRBS, *NR3C1*, and control) in the baseline dataset.

Supplementary File 3. Table of results from the model testing for methylation-dependent regulatory activity in K562 cells at baseline.

Supplementary File 4. Table of enrichment results of mSTARR-seq regulatory regions for ENCODE-defined chromosome states.

Supplementary File 5. Table providing sequencing depth and CpG composition statistics for the baseline dataset and Lea *et al.*, 2018 dataset.

Supplementary File 6. Table of results from the model testing for methylation-dependent regulatory activity in K562 cells at baseline condition using the dataset from (Lea *et al.*, 2018 [↗](#)).

Supplementary File 7. Table of results from the model testing for methylation-dependent regulatory activity in HepG2 cells at baseline.

Supplementary File 8. Table of enrichment results of transcription factor binding motifs for regulatory windows with higher activity in the methylated relative to the sham condition.

Supplementary File 9. Table providing the number of 600 bp windows passing each filter for the baseline, IFNA, dex, and HepG2 datasets.

Supplementary File 10. Table of results from the model testing for methylation-dependent regulatory activity in K562 cells stimulated with IFNA.

Supplementary File 11. Table of results from the model testing for methylation-dependent regulatory activity in K562 cells stimulated with dex.

Supplementary File 12. Table of enrichment results of transcription factor binding motifs for regulatory windows specific to the IFNA treatment (<1% FDR in IFNA and >1% FDR in baseline and Dex).

Supplementary File 13. Table of enrichment results of transcription factor binding motifs for regulatory windows specific to the Dex treatment (<1% FDR in Dex and >1% FDR in baseline and IFNA).

Supplementary File 14. Table of enrichment results of transcription factor binding motifs for regulatory windows specific to the IFNA treatment (<1% FDR in IFNA and >10% FDR in baseline and Dex).

Supplementary File 15. Table of enrichment results of transcription factor binding motifs for regulatory windows specific to the Dex treatment (<1% FDR in Dex and >10% FDR in baseline and IFNA).

Supplementary File 16. Table of results from the model testing for the K562 endogenous gene expression response to IFNA.

Supplementary File 17. Table of Pearson's correlation results, within individuals, between mSTARR-seq enhancer DNA methylation in non-infected macrophages and transcriptional response of nearest genes upon IAV infection.

Supplementary File 18. Table of Pearson's correlation results, within individuals, between mSTARR-seq enhancer DNA methylation in non-infected macrophages and gene expression level in non-infected or IAV-infected macrophages.

Supplementary File 19. Table of Pearson's correlation results, across individuals, between mSTARR-seq enhancer DNA methylation in non-infected macrophages and transcriptional response of nearest genes upon IAV infection.

Supplementary File 20. Table of enrichment results of regulatory activity in early-life adversity associated CpGs relative to regulatory activity in all CpGs on the associated Illumina array.

Supplementary File 21. Library diversity of replicate transformations using the ENCODE Project's PCR bottleneck coefficient (PBC).

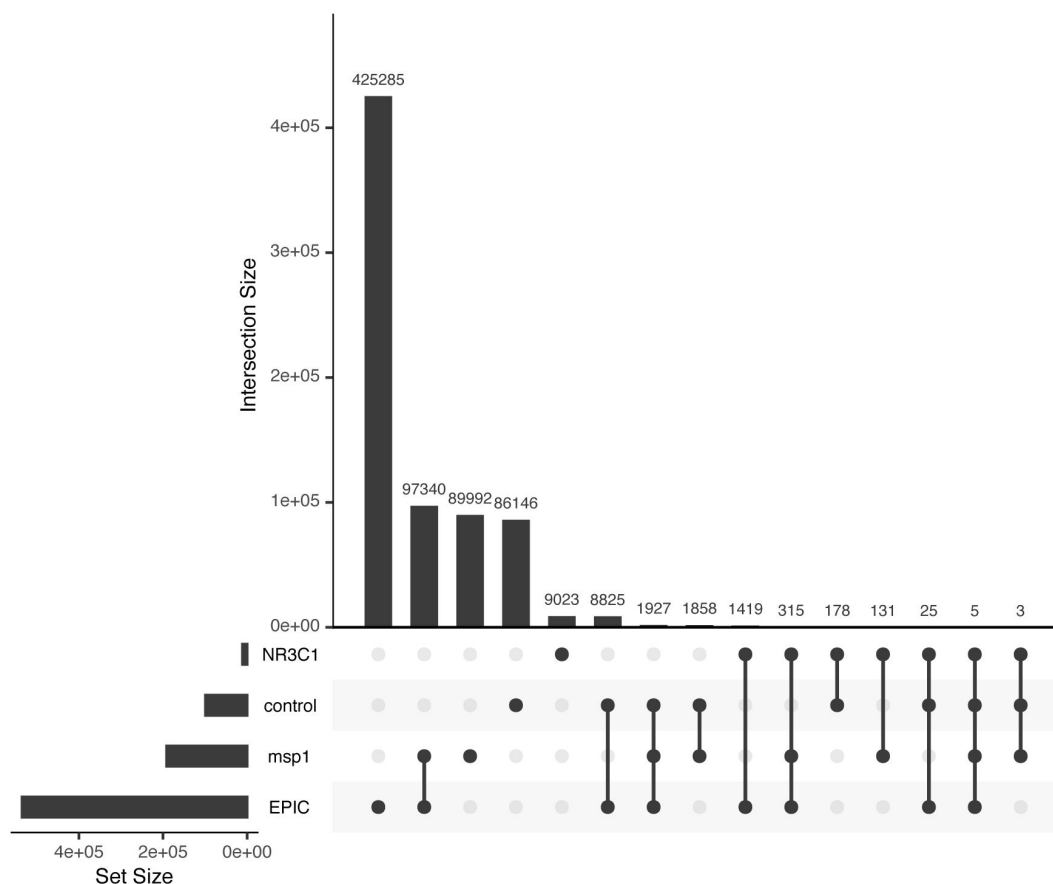


Figure 1-figure supplement 1.

Overlap of target genomic regions with each other.

Upset plot showing the degree to which 600 bp non-overlapping genomic windows are shared between the four target genomic regions (EPIC CpGs, *Msp1* CpG cut sites, the *NR3C1* region, or control sites). Overlap occurs because a single 600 bp genomic window can simultaneously include EPIC CpGs, *Msp1* CpG cut sites, the *NR3C1* region, and/or control sites. This plot includes 722,472 unique windows, reflecting the set of windows containing at least 1 basepair of sequence in the target loci.

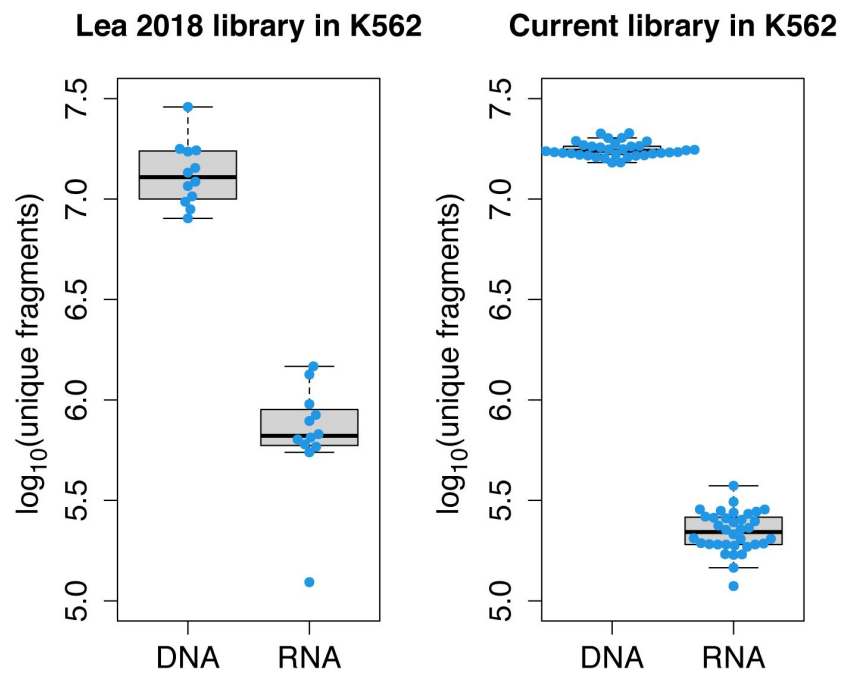


Figure 1-figure supplement 2.

mSTARR-seq library diversity.

Comparison of diversity of unique mSTARR-seq DNA and RNA fragments from the library generated in this study (transfected into K562 cells) relative to the library published in [Lea *et al.*, 2018](#) (independently transfected into K562 cells). Each dot represents an experimental replicate. Each box represents the interquartile range, with the median value depicted as a horizontal bar. Whiskers extend to the most extreme values within 1.5x of the interquartile range.

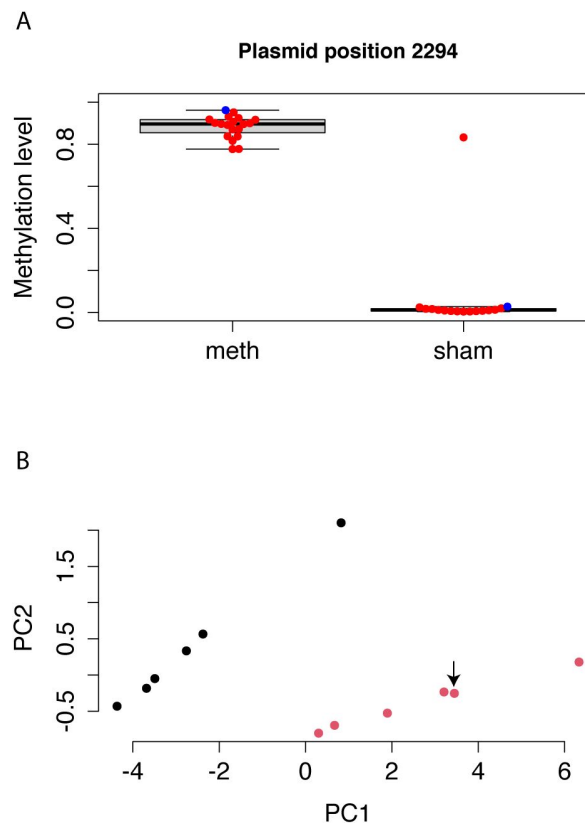


Figure 1-figure supplement 3.

Methylation levels of mSTARR-seq DNA, pre- and post-transfection.

(A) Bisulfite sequencing shows that DNA methylation on the mSTARR-seq plasmid is maintained until the end of the experiment (i.e., 48 hours after transfection), with significantly higher methylation levels in the replicates from the methyltransferase reaction relative to the replicates from the sham methyltransferase reaction (mean methylated = 0.885, mean unmethylated = 0.066; unpaired t-test: $t = -14.66$, $df = 15,124$, $p = 2.39 \times 10^{-10}$). Each dot represents an experimental replicate. Red dots indicate post-transfection DNA samples; the single blue dot per condition indicates pre-transfection DNA methylation levels. Methylation estimates are based on the CpG at the position 2294, which is located in the plasmid region used for Gibson assembly. We assessed methylation of this CpG, rather than across CpGs genome-wide, because the genomic coverage of our bisulfite sequencing data across replicates was too variable to perform reliable site-by-site analysis of DNA methylation levels before and after the 48 hour experiment. One sample from the dex sham reaction, L31395, shows an unexpectedly high level of methylation, which appears to be due to an error during generation of the bisulfite sequencing library (e.g., mislabeled tube or poor bisulfite conversion), and not the experimental replicate of cells itself, as the mSTARR-seq RNA library (L31244) from the same replicate clusters with the unmethylated sham replicates as expected (panel B). **(B)** The first two principal components summarizing overall counts of mSTARR-seq reads for the dex-treated RNA samples (i.e., the raw readout of overall regulatory activity). Each dot represents an experimental replicate, with red and black indicating sham and methylated replicates, respectively. Overall regulatory activity of sample L31244 (indicated by arrow) clusters with the sham replicates as expected, suggesting that this replicate was indeed transfected with sham-treated mSTARR-seq DNA.

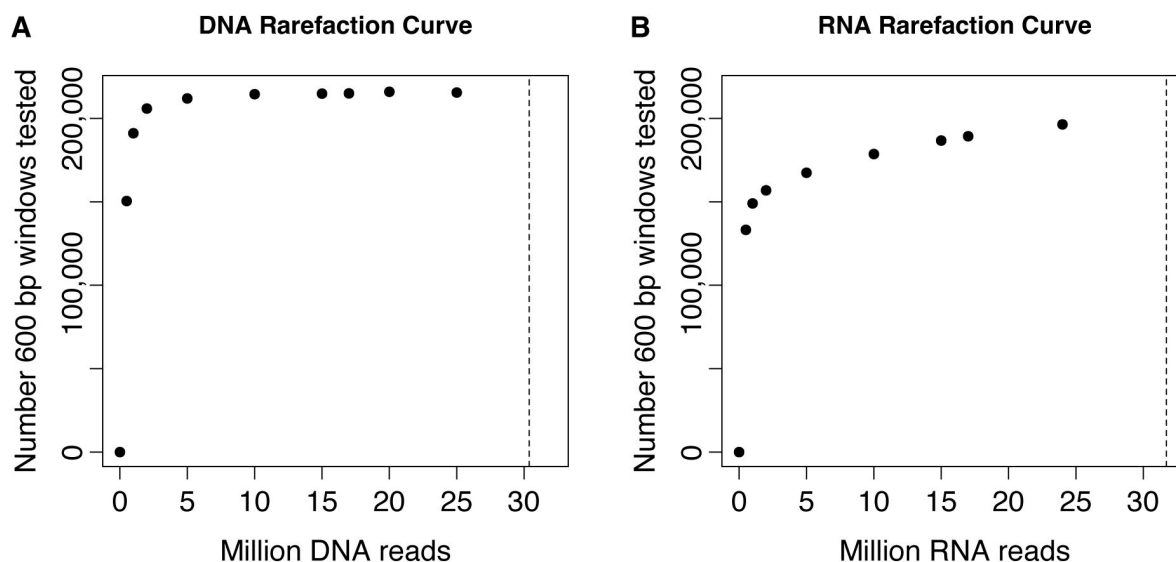


Figure 1-figure supplement 4.

Rarefaction curve showing total number of windows formally tested for regulatory activity, as a function of number of reads sequenced per DNA or RNA replicate.

Sequencing reads from the (A) DNA replicates or (B) RNA replicates of the baseline dataset were rarefied to the values shown on the x-axis before running the data processing steps and applying the filtering criteria described in the Materials and Methods for the full data set. Dashed vertical lines represent the mean number of sequenced reads per DNA replicate (mean [SD] = 30.375 million [3.335 million]) or RNA replicate (mean [SD] = 31.712 million [8.194 million]) in the full baseline dataset. These analyses show that our sequencing effort saturated the number of formally analyzable windows based on either our criteria for inclusion based on DNA library sequencing depth or RNA library sequencing depth.

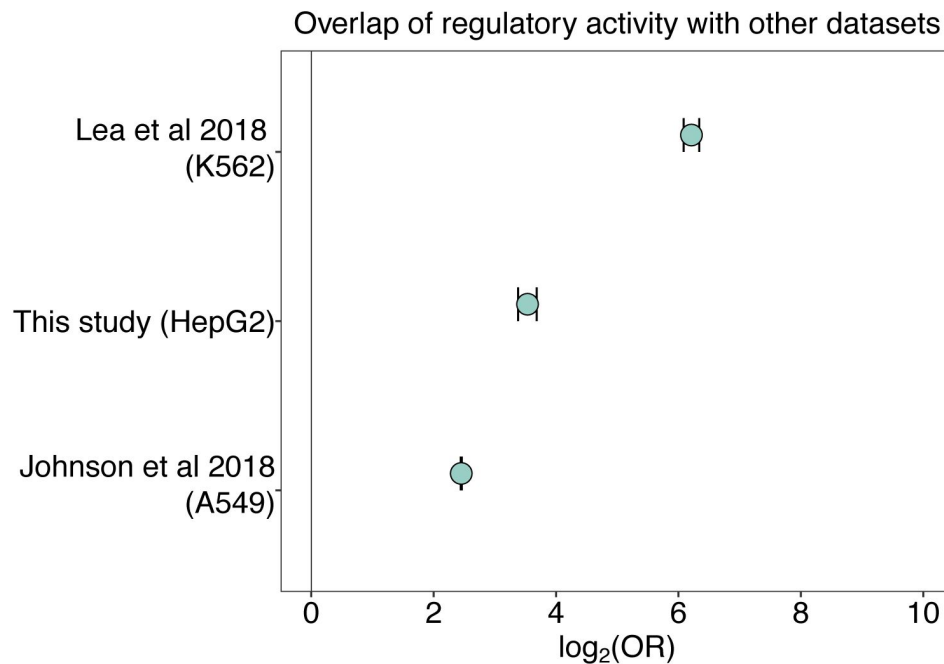


Figure 1-figure supplement 5.

Overlap of regulatory activity across datasets.

Regulatory regions (in either the unmethylated sham condition, the methylated condition, or both) identified via mSTARR-seq in this study significantly overlap with: K562 regulatory regions (in either the unmethylated sham or methylated condition, or both) from a previously generated mSTARR-seq dataset reanalyzed with our pipeline (Lea et al., 2018 [DOI](#)) ($\log_2(\text{OR})$ [95% CI] = 6.212 [6.086, 6.440], $p < 1.0 \times 10^{-300}$); regulatory regions (in either the unmethylated sham or methylated condition, or both) from an mSTARR-seq experiment in HepG2 liver cells ($\log_2(\text{OR})$ [95% CI] = 3.534 [3.381, 3.684], $p = 5.21 \times 10^{-307}$); and regulatory regions from a conventional STARR-seq experiment (i.e., an unmethylated condition) in A549 lung epithelial cells (Johnson et al., 2018 [DOI](#)) ($\log_2(\text{OR})$ [95% CI] = 2.451 [2.442, 2.461], $p < 1.0 \times 10^{-300}$). Bars represent 95% confidence intervals.

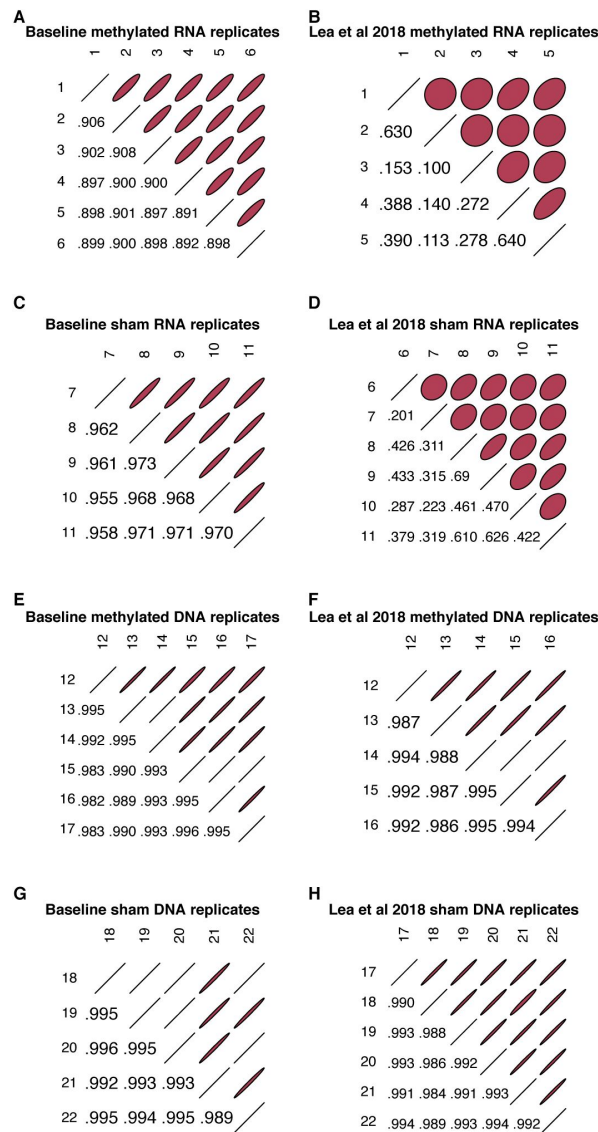


Figure 1-figure supplement 6.

Correlations between RNA and DNA replicates.

Pearson correlations (r) of raw counts between RNA replicates (A-D) and between DNA replicates (E-H) within the windows we formally analyzed for enhancer activity in the baseline dataset reported here and in [Lea et al., 2018](#) [\[1\]](#), following a uniform data processing pipeline. All replicate pairs (both RNA and DNA, in both sham and methylated conditions) in the baseline dataset show correlations ≥ 0.89 , demonstrating replicate reproducibility comparable to other STARR-seq studies (e.g., [Klein et al., 2020](#) [\[2\]](#)). For RNA libraries, replicates in the baseline dataset are more correlated than in the [Lea et al., 2018](#) [\[1\]](#) dataset (RNA replicates: baseline mean $r = 0.926$; [Lea et al., 2018](#) [\[1\]](#) mean $r = 0.347$), although DNA replicates show similar inter-replicate consistency (baseline mean $r = 0.992$; [Lea et al., 2018](#) mean $r = 0.991$).

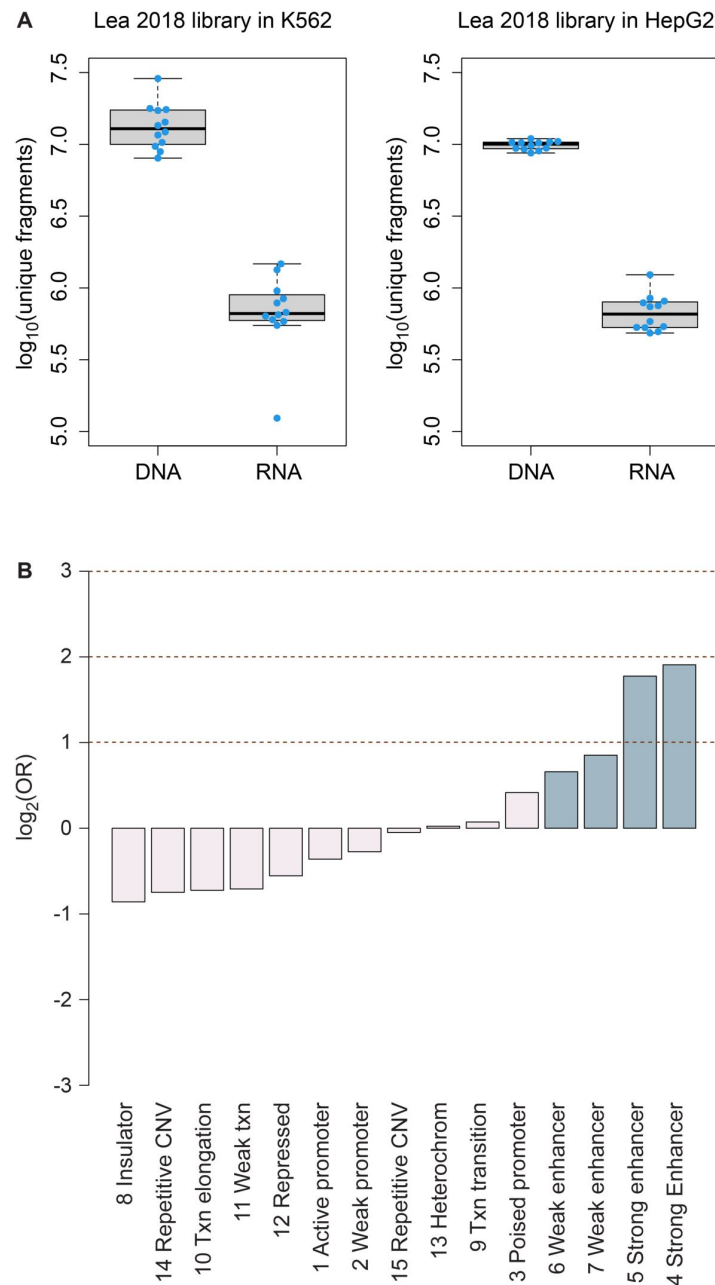


Figure 1-figure supplement 7.

Library diversity and regions of regulatory activity in HepG2 cells.

(A) Comparison of diversity of unique mSTARR-seq DNA and RNA fragments from the library published in [Lea et al., 2018](#) (transfected into K562 cells) versus the same library transfected into HepG2 cells in this study. Each dot represents an experimental replicate. Each box represents the interquartile range, with the median value depicted as a horizontal bar. Whiskers extend to the most extreme values within 1.5x of the interquartile range. **(B)** mSTARR-seq regulatory activity in HepG2 cells is strongly enriched in ENCODE-defined enhancers (indicated in blue) and some classes of promoters, and depleted in repressed and repetitive states.

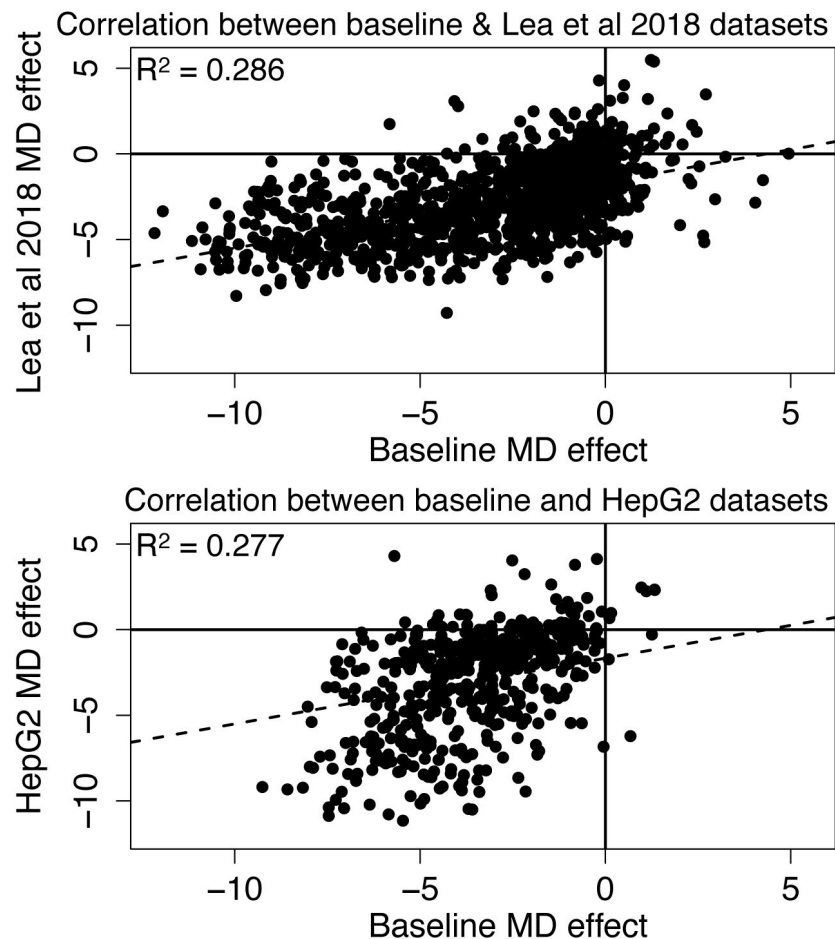


Figure 1-figure supplement 8.

Methylation-dependent regulatory activity across datasets.

Effects of methylation on regulatory activity estimated in this study in the baseline dataset are consistent with methylation effects in K562s estimated from a previously generated mSTARR-seq dataset (Lea *et al.*, 2018 [DOI](#)) and with methylation effects estimated in HepG2 liver cells (Lea *et al.*, 2018 [DOI](#): Pearson's $r = 0.534$ for 1250 windows with $\text{FDR} < 1\%$ in both data sets, $R^2 = 0.286$, $p = 3.19 \times 10^{-93}$; HepG2: Pearson's $r = 0.526$ for 511 windows with $\text{FDR} < 1\%$ in both data sets, $R^2 = 0.277$, $p = 8.87 \times 10^{-38}$). Each dot represents a 600 bp regulatory window identified, in either the sham or methylated states, in both datasets ($\text{FDR} < 1\%$; note that not all regulatory windows show significant methylation dependence [MD]). Dashed lines are the best fit lines. In all cases, negative effect sizes correspond to reduced activity in the methylated condition and positive effect sizes correspond to increased activity in the methylated condition.

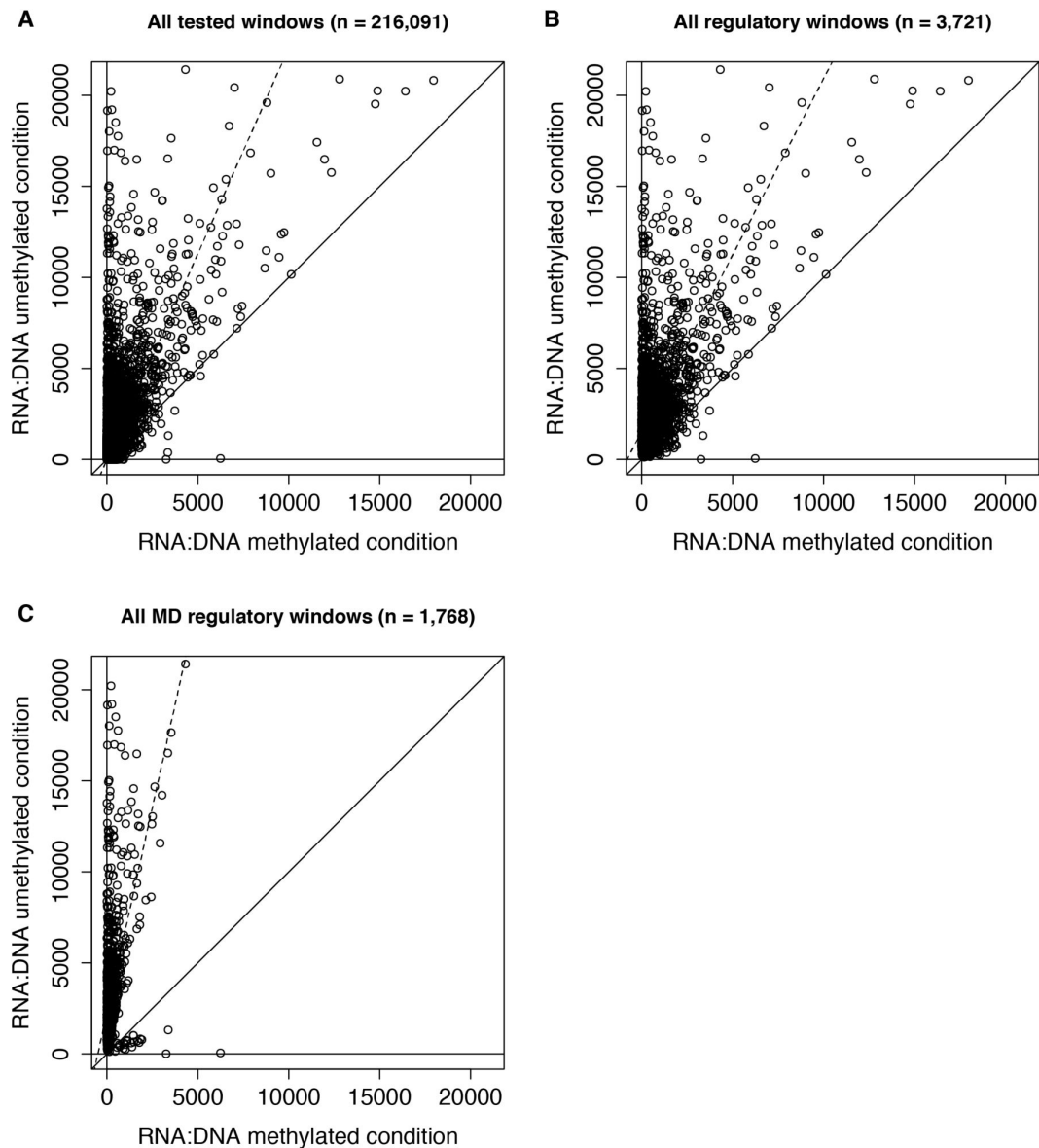


Figure 1-figure supplement 9.

RNA to DNA ratios in the methylated and unmethylated replicates in the baseline dataset.

Mean RNA (in counts per million) to DNA ratios for methylated replicates (x-axis) versus unmethylated replicates (y-axis). A constant of 0.5 was added to the initial raw counts to ensure no denominator values were 0. Each dot represents a 600 bp window that was formally tested for enhancer activity (**A**), exhibited significant regulatory activity (**B**), or exhibited significant methylation-dependent regulatory activity (**C**) in the baseline dataset. Solid diagonal lines represent $y = x$, and dashed lines represent the best fit lines. As expected, 600 bp windows tend to show higher RNA to DNA ratios in the unmethylated condition relative to the methylated condition.

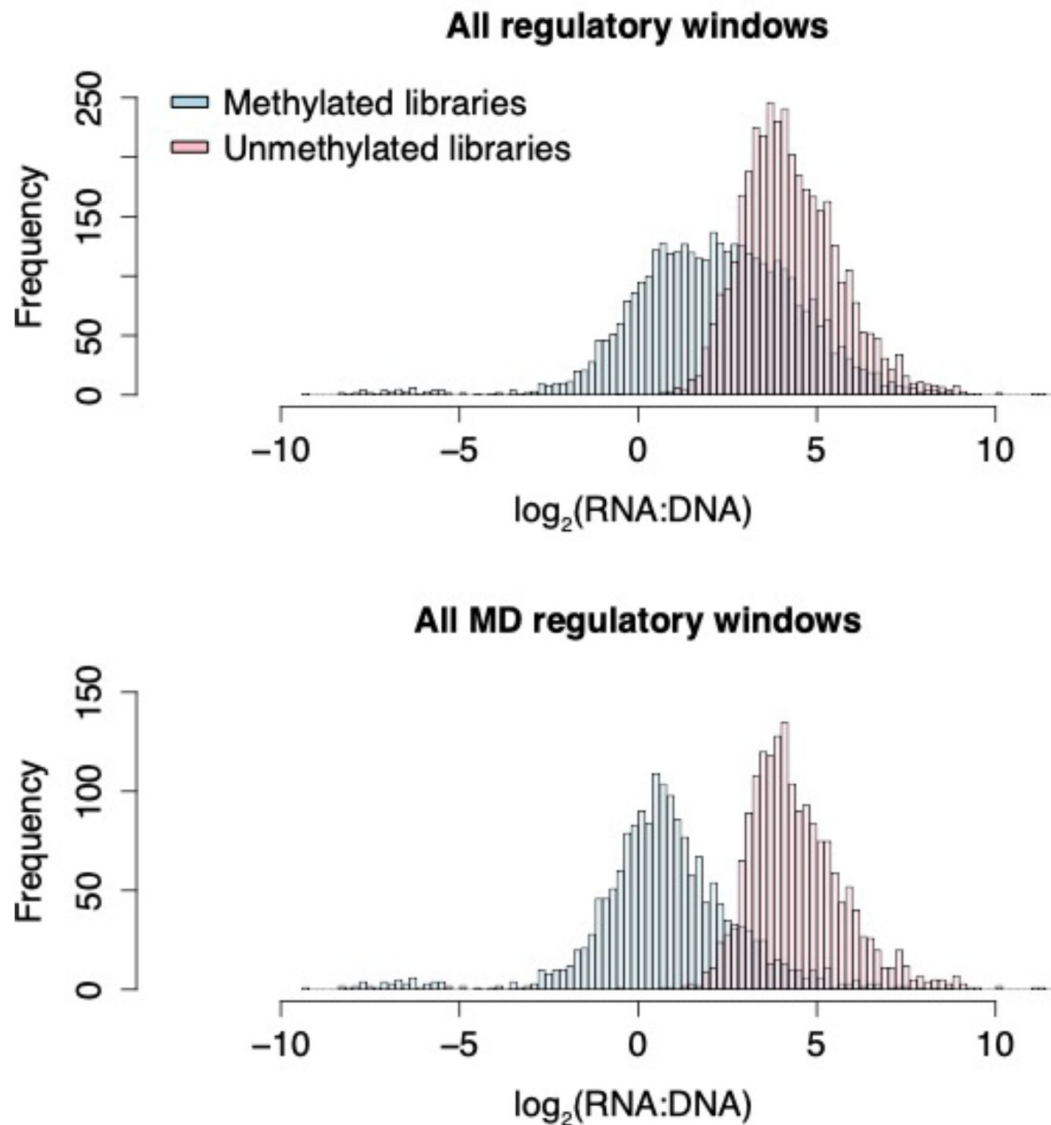


Figure 1-figure supplement 10.

Histograms of RNA to DNA ratios in the baseline dataset.

The x-axis represents the $\log_2(\text{mean RNA [in counts per million] to DNA ratios})$ for the baseline dataset. A constant of 0.5 was added to the initial raw counts to prevent 0 from being in the denominator. Lower values on the x-axis indicate windows showing lower regulatory activity.

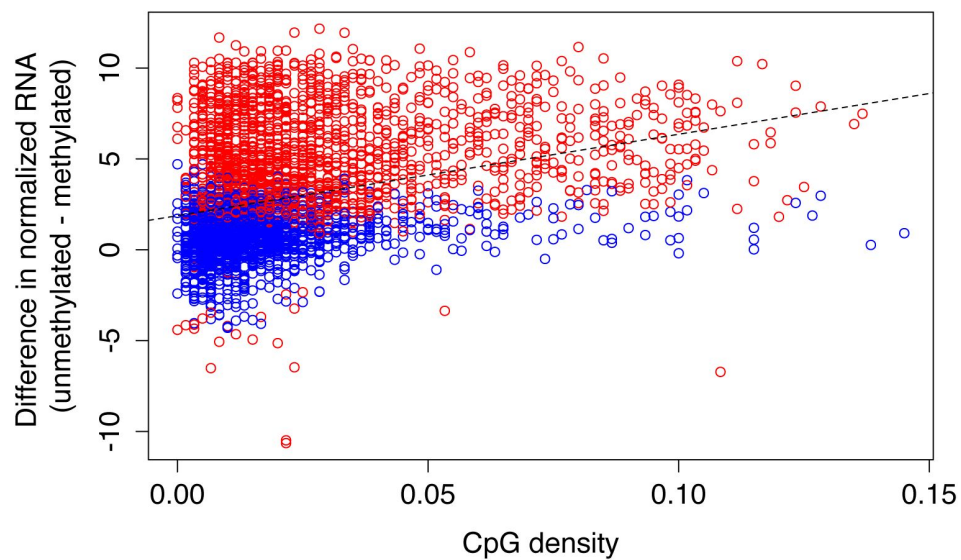


Figure 1-figure supplement 11.

Relationship between CpG density and methylation-dependent regulatory activity.

CpG-dense mSTARR-seq regulatory regions are more likely to be repressed by DNA methylation (positive y-axis value; Spearman's $\rho = 0.370$, $p = 9.865 \times 10^{-121}$; $n = 3,721$ regions with mSTARR-seq regulatory activity). Each dot represents a 600 bp window that showed significant regulatory activity ($FDR < 1\%$). Red and blue dots represent regulatory windows where methylation-dependent activity was or was not detected, respectively. The dashed line represents the best fit line.

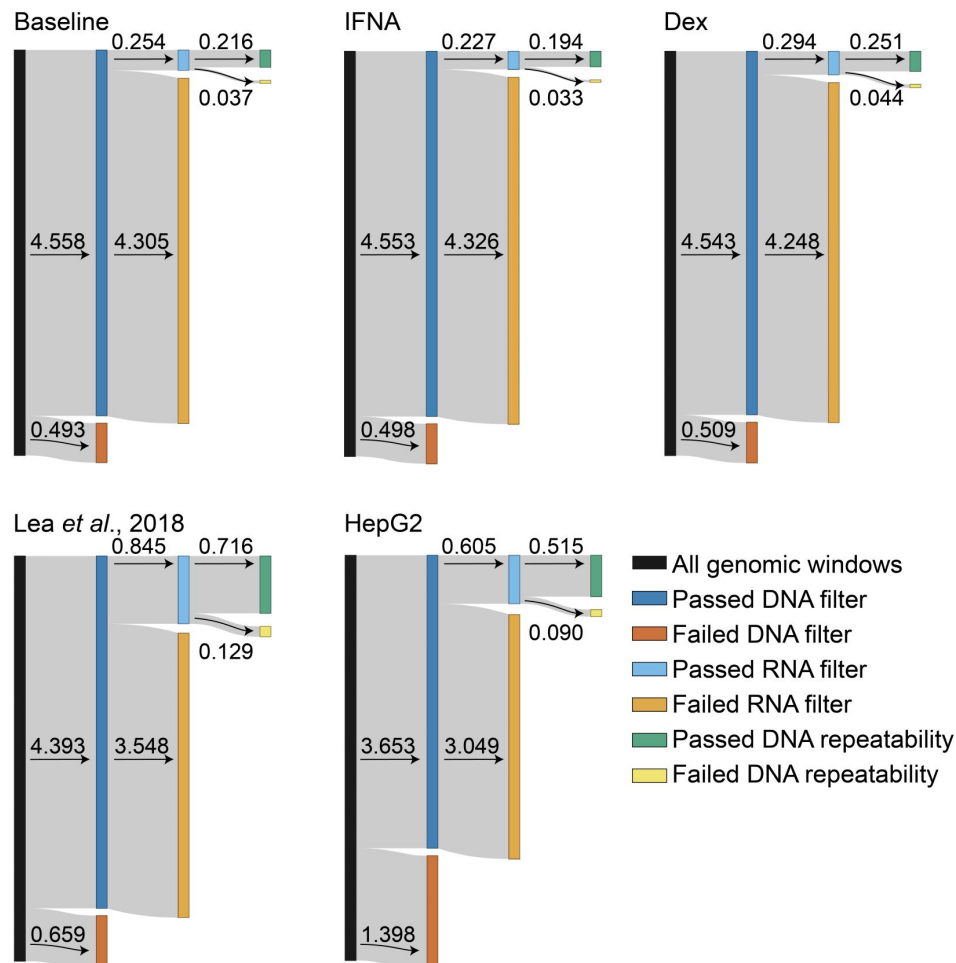


Figure 2-figure supplement 1.

Filtering results across datasets.

Each of the 5 datasets began with 5,051,776 600 bp genomic windows. For each of the 5 datasets, we reduced the dataset to windows that had nonzero counts in at least three DNA samples in the methylated condition *and* three DNA samples in the unmethylated condition (i.e., 6 DNA samples total; “DNA filter”). We then reduced the dataset to windows that had nonzero counts in at least three RNA samples in either the methylated *or* unmethylated condition (“RNA filter”). Finally, we retained only windows that showed high repeatability across DNA samples, following (Lea *et al.*, 2018) (“DNA repeatability”). Numbers correspond to million windows that passed or failed each filter for which the arrow points to. Note that one mSTARR RNA-seq sample in the baseline condition [sample ID L31250] was removed from further analysis because it had an unusually high proportion of zero counts in the testable windows; we therefore also removed the corresponding paired DNA sample prior to analysis). See Supplementary File 9 for the precise window numbers corresponding to the plots.

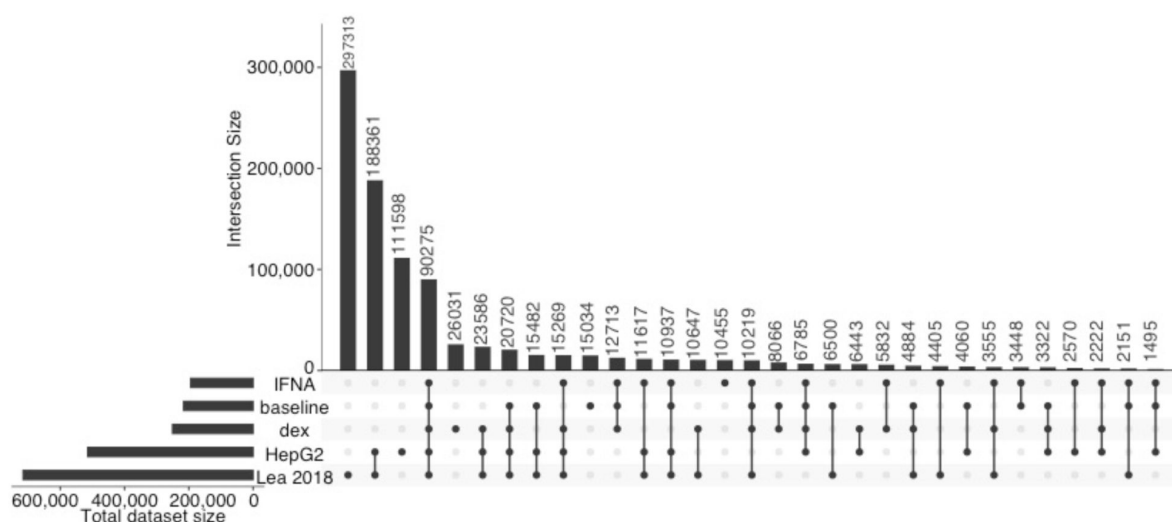


Figure 2-figure supplement 2.

Overlap of tested genomic windows across datasets.

Upset plots showing the degree to which 600 bp non-overlapping genomic windows are shared between five datasets (baseline null, IFNA, dex, HepG2, and Lea *et al.*, 2018; all datasets were analyzed following the same pipeline). Note that the same input library was used in Lea *et al.*, 2018 [Lea et al., 2018](#) and the HepG2 experiment, which differs from the library used here for baseline, dex, and IFNA experiments.

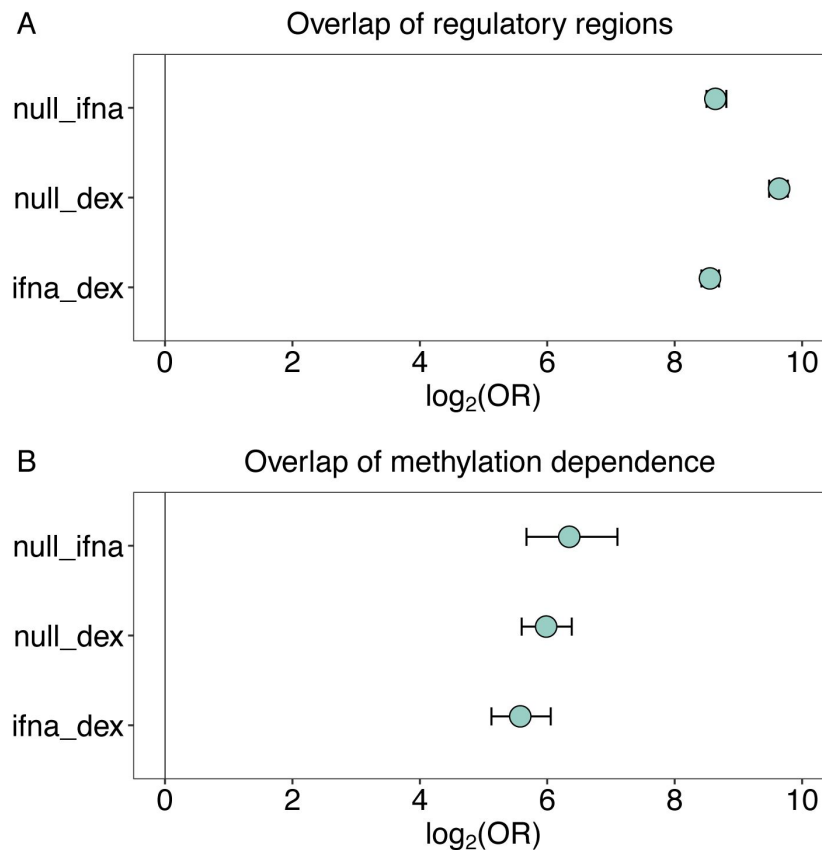


Figure 2-figure supplement 3.

Overlap of regulatory activity and effects of methylation across environmental conditions.

(A) Regulatory regions in the baseline condition are highly likely to retain regulatory activity upon challenge with IFNA or dex (IFNA $\log_2(\text{OR})$ [95% CI] = 8.639 [8.499, 8.812], $p < 1.0 \times 10^{-300}$; dex $\log_2(\text{OR})$ [95% CI] = 9.640 [9.483, 9.776], $p < 1.0 \times 10^{-300}$). Regulatory regions also significantly overlap between IFNA- and dex-challenged cells ($\log_2(\text{OR})$ [95% CI] = 8.554 [8.420, 8.698], $p < 1.0 \times 10^{-300}$). **(B)** Regulatory windows identified in two environmental conditions tend to share significant effects of DNA methylation on regulatory activity (i.e., interaction effects between methylation and regulatory activity) across the two environmental conditions (baseline and IFNA $\log_2(\text{OR})$ [95% CI] = 6.345 [5.673, 7.102], $p < 7.33 \times 10^{-239}$; baseline and dex $\log_2(\text{OR})$ [95% CI] = 5.982 [5.599, 6.384], $p < 1.0 \times 10^{-300}$; IFNA and dex $\log_2(\text{OR})$ [95% CI] = 5.576 [5.123, 6.055], $p < 4.70 \times 10^{-266}$).

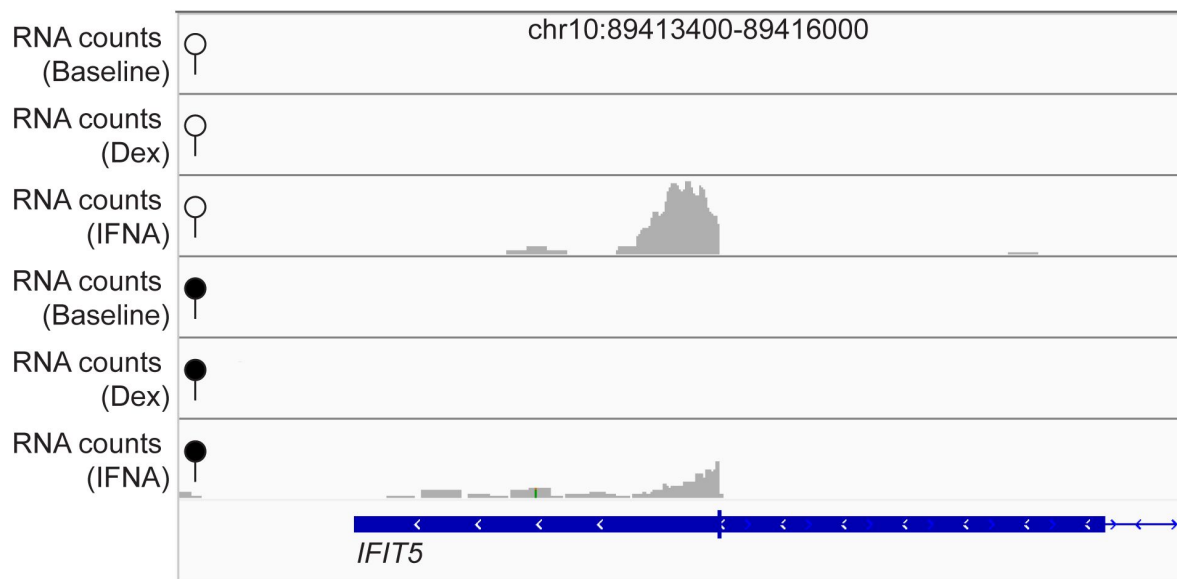


Figure 2-figure supplement 4.

***IFIT5* endogenous gene expression is responsive to IFNA stimulation.**

Tracks show non-normalized, raw read pile-ups of endogenous *IFIT5* (ENSG00000152778) gene expression in either the unmethylated (open circle) or methylated (filled circle) condition, with all y-axis maximums set to 100. The replicates for endogenous gene expression shown here are the same replicates used for measuring regulatory activity shown in **Figure 2A**. *IFIT5* is only detectably expressed after IFNA stimulation (note that the difference in peak heights between the IFNA-stimulated unmethylated and methylated conditions is because the plot shows raw reads; there is no effect of methylation treatment on endogenous *IFIT5* gene expression after normalization for library size: $p = 0.489$).

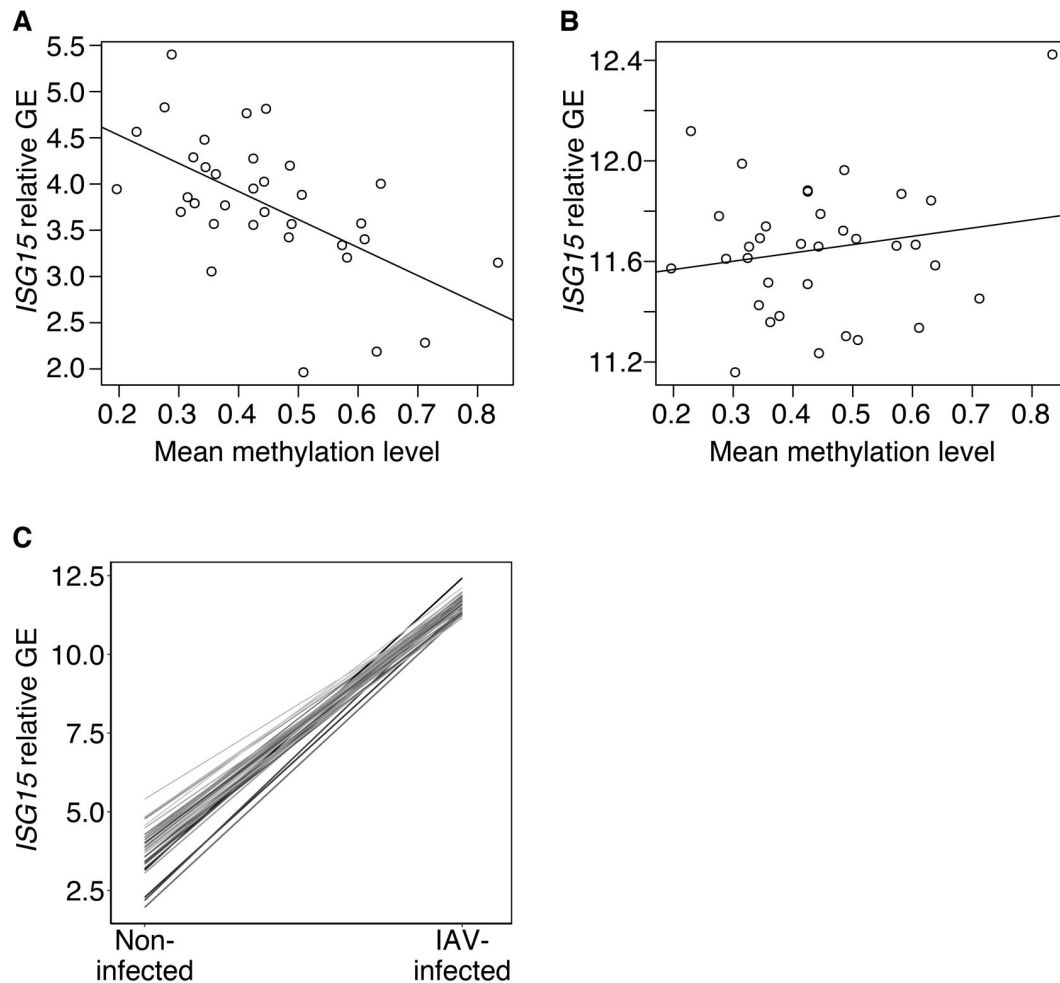


Figure 3-figure supplement 1.

Across individuals, methylation in the mSTARR-seq annotated enhancer chr1:1013400-1014000 predicts the *ISG15* gene expression response to flu.

(A) Across individuals, average methylation within the mSTARR-seq annotated enhancer chr1:1013400-1014000 in non-infected baseline macrophages significantly predicts *ISG15* gene expression (GE) in the non-infected condition ($R^2 = 0.324$, $p = 2.66 \times 10^{-4}$), but (B) not in the IAV-infected condition ($R^2 = 0.001$, $p = 0.316$). Each dot represents an individual. Relative GE is $\log(\text{CPM})$ after regressing out the effects of sequencing batch, age, and admixture. (C) Individuals with relatively low methylation in the mSTARR-seq chr1:1013400-1014000 enhancer (indicated by lighter line color) in the baseline, non-infected condition tend to have higher *ISG15* gene expression in the non-infected condition, ultimately resulting in a shallower *ISG15* transcriptional response to IAV infection (as indicated by slope of the line). Each line represents an individual.

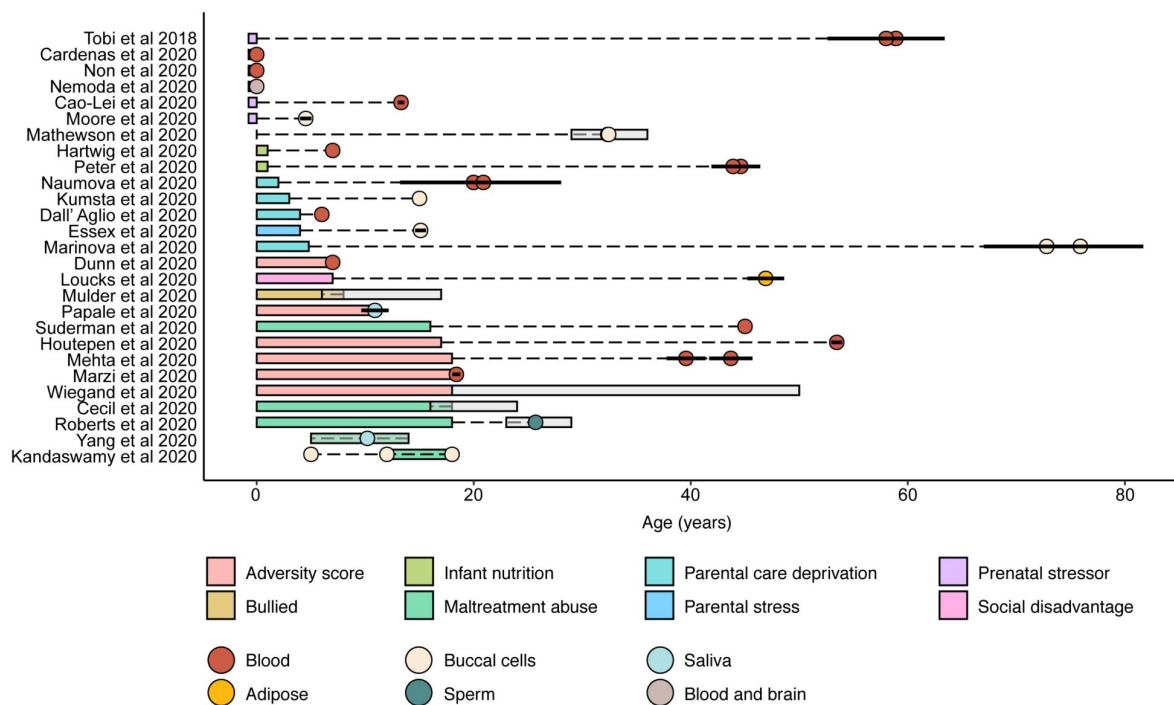


Figure 4-figure supplement 1.

Summary of early-life adversity (ELA) studies.

Colored rectangles indicate ranges of age at adversity. Circles represent mean ages at sample collection and are colored according to the tissue type used to measure methylation levels. Ranges and standard errors of ages at sample collection, when reported, are indicated by gray rectangles and horizontal black solid lines, respectively. See Supplementary File 20 for results of Fisher's exact tests assessing enrichment of ELA-associated CpGs for regulatory activity.

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Reviewer #1 (Public Review):

In this manuscript, the authors explore the effects of DNA methylation on the strength of regulatory activity using massively parallel reporter assays in cell lines on a genome-wide level. This is a follow-up of their first paper from 2018 that describes this method for the first time. In addition to adding more in depth information on sequences that are explored by many researchers using two main methods, reduced bisulfite sequencing and sites represented on the Illumina EPIC array, they now show also that DNA methylation can influence changes in regulatory activity following a specific stimulation, even in absence of baseline effects of DNA methylation on activity. In this manuscript, the authors explore the effects of DNA methylation on the response to Interferon alpha (INFA) and a glucocorticoid receptor agonist (dexamethasone). The authors validate their baseline findings using additional datasets, including RNAseq data and show convergences across two cell lines. The authors then map the methylation x environmental challenge (IFNA and dex) sequences identified *in vitro* to explore whether their methylation status is also predictive of regulatory activity *in vivo*. This is very convincingly shown for INFA response sequences, where baseline methylation is predictive of the transcriptional response to flu infection in human macrophages, an infection that triggers the INF pathways. The extension of the functional validity of the dex-response altering sequences is less convincing. Sequences altering the response to glucocorticoids, however, were not enriched in DNA methylation sites associated with exposure to early adversity which the authors interpret that "they are not links on the causal pathway between early life disadvantage and later life health outcomes, but rather passive biomarkers. However, this approach does not seem an optimal model to explore this relationship *in vivo*. This is because exposure to early adversity and its consequences is not directly correlated with glucocorticoid release and changes in DNA methylation levels following early adversity could be related to many physiological mechanisms, and overall, large datasets and meta-analyses do not show robust associations of exposure to early adversity and DNA methylation changes. Here other datasets, such as from Cushing patients maybe of more interest.

After revision, the authors have now discussed this issue carefully, so that this point is addressed.

Overall, the authors provide a great resource of DNA methylation sensitive enhancers that can now be used for functional interpretation of large scale datasets (that are widely generated in the research community), given the focus on sites included in RBSS and the Illumina EPIC array. In addition, their data lends support that difference in DNA methylation can alter responses to environmental stimuli and thus of the possibility that environmental exposures that alter DNA methylation can also alter subsequent response to this exposure, in line with the theory of epigenetic embedding of prior stimuli/experiences. The conclusions related to the early adversity data should be reconsidered in light of the comments above.

<https://doi.org/10.7554/eLife.89371.2.sa1>

Reviewer #2 (Public Review):

This work presents a remarkably extensive set of experiments, assaying the interaction between methylation and expression across most CpG positions in the genome in two cell types. To this end, the authors use mSTARR-seq, a high-throughput method, which they have previously developed, where sequences are tested for their regulatory activity in two conditions (methylated and unmethylated) using a reporter gene. The authors use these data to study two aspects of DNA methylation: 1. Its effect on expression, and 2. Its interaction with the environment. Overall, they identify a small number of 600 bp windows that show

regulatory potential, and a relatively large fraction of these show an effect of methylation on expression. In addition, the authors find regions exhibiting methylation-dependent response to two environmental stimuli (interferon alpha and glucocorticoid dexamethasone).

The questions the authors address represent some of the most central in functional genomics, and the method utilized is currently the best method to do so. The scope of this study is very impressive and I am certain that these data will become an important resource for the community. The authors are also able to report several important findings, including that pre-existing DNA methylation patterns can influence the response to subsequent environmental exposures.

<https://doi.org/10.7554/eLife.89371.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

In this manuscript, the authors explore the effects of DNA methylation on the strength of regulatory activity using massively parallel reporter assays in cell lines on a genome-wide level. This is a follow-up of their first paper from 2018 that describes this method for the first time. In addition to adding more indepth information on sequences that are explored by many researchers using two main methods, reduced bisulfite sequencing and sites represented on the Illumina EPIC array, they now show also that DNA methylation can influence changes in regulatory activity following a specific stimulation, even in absence of baseline effects of DNA methylation on activity. In this manuscript, the authors explore the effects of DNA methylation on the response to Interferon alpha (IFNA) and a glucocorticoid receptor agonist (dexamethasone). The authors validate their baseline findings using additional datasets, including RNAseq data, and show convergences across two cell lines. The authors then map the methylation x environmental challenge (IFNA and dex) sequences identified in vitro to explore whether their methylation status is also predictive of regulatory activity in vivo. This is very convincingly shown for IFNA response sequences, where baseline methylation is predictive of the transcriptional response to flu infection in human macrophages, an infection that triggers the INF pathways.

Thank you for your strong assessment of our work!

The extension of the functional validity of the dex-response altering sequences is less convincing.

We agree. We note that genes close to dex-specific mSTARR-seq enhancers tend to be more strongly upregulated after dex stimulation than those near shared enhancers, which parallels our results for IFNA (lines 341-344). However, there is unfortunately no comparable data set to the human flu data set (i.e., with population-based whole genome-bisulfite sequencing data before and after dex challenge), so we could not perform a parallel in vivo validation step. We have added this caveat to the revised manuscript (lines 555-557).

Sequences altering the response to glucocorticoids, however, were not enriched in DNA methylation sites associated with exposure to early adversity. The authors interpret that "they are not links on the causal pathway between early life disadvantage and later life health outcomes, but rather passive biomarkers". However, this approach does not seem an optimal model to explore this relationship in vivo. This is because exposure to early

adversity and its consequences is not directly correlated with glucocorticoid release and changes in DNA methylation levels following early adversity could be related to many physiological mechanisms, and overall, large datasets and meta-analyses do not show robust associations of exposure to early adversity and DNA methylation changes. Here, other datasets, such as from Cushing patients may be of more interest.

Thank you for making these important points. We have expanded the set of caveats regarding the lack of enrichment of early adversity-reported sites in the mSTARR-data set (lines 527-533). Specifically, we note that the relationship between early adversity and glucocorticoid physiology is complex (e.g., Eisenberger and Cole, 2012; Koss and Gunnar, 2018) and that dex challenge models one aspect of glucocorticoid signaling but not others (e.g., glucocorticoid resistance). Nevertheless, we also see little evidence for enrichment of early adversity-associated sites in the mSTARR data set at baseline, independently of the dex challenge experiment (lines 483-485; Figure 4).

We also agree that large data sets (e.g., Houtepen et al., 2018; Marzi et al., 2018) and reviews (e.g., Cecil et al., 2020) of early adversity and DNA methylation in humans show limited evidence of associations between early adversity and DNA methylation levels. However, the idea that early adversity impacts downstream outcomes remains pervasive in the literature and popular science (see Dubois et al., 2019), which we believe makes tests like ours important to pursue. We also hope that our data set (and others generated through these methods) will be useful in interpreting other settings in which differential methylation is of interest as well—in line with your comment below. We have clarified both of these points in the revised manuscript (lines 520-522; 536-539).

Overall, the authors provide a great resource of DNA methylation-sensitive enhancers that can now be used for functional interpretation of large-scale datasets (that are widely generated in the research community), given the focus on sites included in RBSS and the Illumina EPIC array. In addition, their data lends support that differences in DNA methylation can alter responses to environmental stimuli and thus of the possibility that environmental exposures that alter DNS methylation can also alter the subsequent response to this exposure, in line with the theory of epigenetic embedding of prior stimuli/experiences. The conclusions related to the early adversity data should be reconsidered in light of the comments above.

Thank you! And yes, we have revised our discussion of early life adversity effects as discussed above.

Reviewer #1 (Recommendations For The Authors):

While the paper has a lot of strengths and provides new insight into the epigenomic regulation of enhancers as well as being a great resource, there are some aspects that would benefit from clarification.

a. It would be great to have a clearer description of how many sequences are actually passing QC in the different datasets and what the respective overlaps are in bps or 600bp windows. Now often only % are given. Maybe a table/Venn diagram for overview of the experiments and assessed sequences would help here. This concern the different experiments in the K652, A549, and Hep2G cell lines, including stimulations.

We now provide a supplementary figure and supplementary table providing, for each dataset, the number of 600 bp windows passing each filter (Figure 2-figure supplement 1; Supplementary File 9), as well as a supplementary figure providing an upset plot to show the number of assessed sequences shared across the experiments (Figure 2-figure supplement 2).

b. It would also be helpful to have a brief description of the main differences in assessed sequences and their coverage of the old (2018) and new libraries in the main text to be able better interpret the validation experiments.

We now provide information on the following characteristics for the 2018 data set versus the data set presented for the first time here: mean ($\pm$ SD) number of CpGs per fragment; mean ($\pm$ SD) DNA sequencing depth; and mean ($\pm$ SD) RNA sequencing depth (lines 169-170 provide values for the new data set; in line 194, we reference Supplementary File 5, which provides the same values for the old data set). Notably, the coverage characteristics of analyzed windows in both data sets are quite high (mean DNA-seq read coverage = 94x and mean RNA-seq read coverage = 165x in the new data set at baseline; mean DNA-seq read coverage = 22x and mean RNA-seq read coverage = 54x in Lea et al. 2018).

c. Statements of genome-wide analyses in the abstract and discussion should be a bit tempered, as quite a number of tested sites do not pass QC and do not enter the analysis. From the results it seems like from over 4.5 million sequences, only 200,000 are entering the analysis.

The reason why many of the windows are not taken forward into our formal modeling analysis is that they fail our filter for RNA reads because they are never (or almost never) transcribed—not because there was no opportunity for transcription (i.e., the region was indeed assessed in our DNA library, and did not show output transcription, as now shown in Figure 2-figure supplement 1). We have added a rarefaction analysis (lines 715-722 in Materials and Methods) of the DNA fragment reads to the revised manuscript which supports this point. Specifically, it shows that we are saturated for representation of unique genomic windows (i.e., we are above the stage in the curve where the proportion of active windows would increase with more sequencing; Figure 1-figure supplement 4). Similarly, a parallel rarefaction curve for the mSTARR-seq RNA-seq data (Figure 1-figure supplement 4) shows that we would gain minimal additional evidence for regulatory activity with more sequencing depth. We now reference these analyses in revised lines 179-184 and point to the supporting figure in line 182.

In other words, our analysis is truly genome-wide, based on the input sequences we tested. Most of the genome just doesn't have regulatory activity in this assay, despite the potential for it to be detected given that the relevant sequences were successfully transfected into the cells.

d. Could the authors comment on the validity of the analysis if only one copy is present (cut-off for QC)?

We think this question reflects a misunderstanding of our filtering criteria due to lack of clarity on our part, which we have modified in the revision. We now specify that the mean DNA-seq sequencing depth per sample for the windows we subjected to formal modeling was quite high:

93.91 $\pm$ 10.09 SD (range = 74.5 – 113.5x) (see revised lines 169-170). In other words, we never analyze windows in which there is scant evidence that plasmids containing the relevant sequence were successfully transfected (lines 170-172).

Our minimal RNA-seq criteria require non-zero counts in at least 3 replicate samples within either the methylated condition or the unmethylated condition, or both (lines 166-168). Because we know that multiple plasmids containing the corresponding sequence are present for all of these windows—even those that just cross the minimal RNA-seq filtering threshold—we believe our results provide valid evidence that all analyzed windows present the opportunity to detect enhancer activity, but many do not act as enhancers (i.e., do not result

in transcribed RNA). Notably, we observe a negligible correlation between DNA sequencing depth for a fragment, among analyzed windows, and mSTARR-seq enhancer activity ($R^2 = 0.029$; now reported in lines 183-184). We also now report reproducibility between replicates, in which all replicate pairs have $r > 0.89$, on par with previously published STARR-seq datasets (e.g., Klein et al., 2020; Figure 1-figure supplement 6, pointed to in line 193).

e. While the authors state that almost all of the control sequences contain CpGs sites, could the authors also give information on the total number of CpG sites in the different subsets? Was the number of CpGs in a 600 bp window related to the effects of DNA methylation on enhancer activity?

We now provide the number of CpG sites per window in the different subsets in lines 282-284. As expected, they are higher for EPIC array sites and for RRBS sites because the EPIC array is biased towards CpG-rich promoter regions, and the enzyme typically used in the starting step of RRBS digests DNA at CpG motifs (but control sequences still contain an average of ~13 CpG sites per fragment). We also now model the magnitude of the effects of DNA methylation on regulatory activity as a function of number of CpG sites within the 600 bp windows. Consistent with our previous work in Lea et al., 2018, we find that mSTARR-seq enhancers with more CpGs tend to be repressed by DNA methylation (now reported in lines 216-219 and Figure 1figure supplement 11).

f. In the discussion, a statement on the underrepresented regions, likely regulatory elements with lower CG content, that nonetheless can be highly relevant for gene regulation would be important to put the data in perspective.

Thanks for this suggestion. We agree that regulatory regions, independent of CpG methylation, can be highly relevant, and now clarify in the main text that the “unmethylated” condition of mSTARR-seq is essentially akin to a conventional STARR-seq experiment, in that it assesses regulatory activity regardless of CpG content or methylation status (lines 128-130).

Consequently, our study is well-designed to detect enhancer-like activity, even in windows with low GC content. We now show with additional analyses that we generated adequate DNA-seq coverage on the transfected plasmids to analyze 90.2% of the human genome, including target regions with no or low CpG content (lines 148-149; 153-156; Supplementary file 2). As noted above, we also now clarify that regions dropped out of our formal analysis because we had little to no evidence that any transcription was occurring at those loci, not because sequences for those regions were not successfully transfected into cells (see responses above and new Figure 1-figure supplement 4 and Figure 2-figure supplement 1).

g. To control for differences in methylation of the two libraries, the authors sequence a single CpGs in the vector. Could the authors look at DNA methylation of the 600 bp windows at the end of the experiment, could DNA methylation of these windows be differently affected according to sequence? 48 hours could be enough for de-methylation or re-methylation.

We agree that variation in demethylation or remethylation depending on fragment sequence is possible. We now state this caveat in the main text (lines 158-159), and specify that genomic coverage of our bisulfite sequencing data across replicates are (unfortunately) too variable to perform reliable site-by-site analysis of DNA methylation levels before and after the 48 hour experiment (lines 1182-1185). Instead, we focus on a CpG site contained in the adapter sequence (and thus included in all plasmids) to generate a global estimate of per replicate methylation levels. We also now note that any de-methylation or re-methylation would reduce our power to detect methylation-dependent activity, rather than leading to false positives (lines 163-165).

h. The section on the method for correction for multiple testing should be more detailed as it is very difficult to follow. Why were only 100 permutations used, the empirical p-value could then only be <0.01 ? The description of a subsample of the N windows with positive Betas is unclear, should the permutation not include the actual values and thus all windows - or were there no negative Betas? Was FDR accounting for all elements and pairs?

We have now expanded the text in the Materials and Methods section to clarify the FDR calculation (lines 691, 695-699, 702, 706). We clarify that the 100 permutations were used to generate a null distribution of p-values for the data set (e.g., 100 x 17,461 p-values for the baseline data set), which we used to derive a false discovery rate. Because we base our evidence on FDRs, we therefore compare the distribution of observed p-values to the distribution of p-values obtained via permutation; we do not calculate individual p-values by comparing an observed test statistic against the test statistics for permuted data for that individual window.

We compare the data to permutations with only positive betas because in the observed data, we observe many negative betas. These correspond to windows which have no regulatory activity (i.e., they have many more input DNA reads than RNA-seq reads) and thus have very small p-values in a model testing for DNA-RNA abundance differences. However, we are interested in controlling the false discovery rate of windows that do have regulatory activity (positive betas). In the permuted data, by contrast and because of the randomization we impose, test statistics are centered around 0 and essentially symmetrical (approximately equally likely to be positive or negative). Retaining all p-values to construct the null therefore leads to highly miscalibrated false discovery rates because the distribution of observed values is skewed towards smaller values—because of windows with “significantly” no regulatory activity—compared to the permuted data. We address that problem by using only positive betas from the permutations.

i. The interpretation of the overlap of Dex-response windows with CpGs sites associated with early adversity should be revisited according to the points also mentioned in the public review and the authors may want to consider exploring additional datasets with other challenges.

Thank you, see our responses to the public review above and our revisions in lines (lines 555-559). We agree that comparisons with more data sets and generation of more mSTARR-seq data in other challenge conditions would be of interest. While beyond the scope of this manuscript, we hope the resource we have developed and our methods set the stage for just such analyses.

Reviewer #2 (Public Review):

This work presents a remarkably extensive set of experiments, assaying the interaction between methylation and expression across most CpG positions in the genome in two cell types. To this end, the authors use mSTARR-seq, a high-throughput method, which they have previously developed, where sequences are tested for their regulatory activity in two conditions (methylated and unmethylated) using a reporter gene. The authors use these data to study two aspects of DNA methylation:

1. Its effect on expression, and 2. Its interaction with the environment. Overall, they identify a small number of 600 bp windows that show regulatory potential, and a relatively large fraction of these show an effect of methylation on expression. In addition, the authors find regions exhibiting methylation-dependent responses to two environmental stimuli (interferon alpha and glucocorticoid dexamethasone).

The questions the authors address represent some of the most central in functional genomics, and the method utilized is currently the best method to do so. The scope of this study is very impressive and I am certain that these data will become an important resource for the community. The authors are also able to report several important findings, including that pre-existing DNA methylation patterns can influence the response to subsequent environmental exposures.

Thank you for this generous summary!

The main weaknesses of the study are: 1. The large number of regions tested seems to have come at the expense of the depth of coverage per region (1 DNA read per region per replicate). I have not been convinced that the study has sufficient statistical power to detect regulatory activity, and differential regulatory activity to the extent needed. This is likely reflected in the extremely low number of regions showing significant activity.

We apologize for our lack of clarity in the previous version of the manuscript. Nonzero coverage for half the plasmid-derived DNA-seq replicates is a minimum criterion, but for the baseline dataset, the mean depth of DNA coverage per replicate for windows passing the DNA filter is quite high: 12.723 ± 41.696 s.d. overall, and 93.907 ± 10.091 s.d. in the windows we subjected to full analysis (i.e., windows that also passed the RNA read filter). We now provide these summary statistics in lines 148-149 and 169-170 and Supplementary file 5 (see also our responses to Reviewer 1 above). We also now show, using a rarefaction analysis, that our data set saturates the ability to detect regulatory windows based on DNA and RNA sequencing depth (new Figure 1-figure supplement 4; lines 179-184; 715-722).

1. Due to the position of the tested sequence at the 3' end of the construct, the mSTARR-seq approach cannot detect the effect of methylation on promoter activity, which is perhaps the most central role of methylation in gene regulation, and where the link between methylation and expression is the strongest. This limitation is evident in Fig. 1C and Figure 1-figure supplement 5C, where even active promoters have activity lower than 1. Considering these two points, I suspect that most effects of methylation on expression have been missed.

Thank you for pointing this out. We agree that we have not exhaustively detected methylation-dependent activity in all promoter regions, given that not all promoter regions are active in STARR-seq. However, there is good evidence that some promoter regions can function like enhancers and thus be detected in STARR-seq-type assays (Klein et al., 2020). This important point is now noted in lines 187-189; an example promoter showing methylation-dependent regulatory activity in our dataset is shown in Figure 3E.

We also now clarify that Figure 1C shows significant enrichment of regulatory activity in windows that overlap promoter sequence (line 239). The y-axis is not a measure of activity, but rather the log-transformed odds ratio, with positive values corresponding to overrepresentation of promoter sequences in regions of mSTARR-seq regulatory activity. Active promoters are 1.640 times more likely to be detected with regulatory activity than expected by chance ($p = 1.560 \times 10^{-18}$), which we now report in a table that presents enrichment statistics for all ENCODE elements shown in Figure 1C for clarity (Supplementary

file 4). Moreover, 74.1% of active promoters that show regulatory activity have methylation-dependent activity, also now reported in Supplementary file 4.

Overall, the combination of an extensive resource addressing key questions in functional genomics, together with the findings regarding the relationship between methylation and environmental stimuli makes this a key study in the field of DNA methylation.

Thank you again for the positive assessment!

Reviewer #2 (Recommendations For The Authors):

I suggest the authors conduct several tests to estimate and/or increase the power of the study:

- 1. To estimate the potential contribution of additional sequencing depth, I suggest the authors conduct a downsampling analysis. If the results are not saturated (e.g., the number of active windows is not saturated or the number of differentially active windows is not saturated), then additional sequencing is called for.*

We appreciate the suggestion. We have now performed a downsampling/rarefaction curve analysis in which we downsampled the number of DNA reads, and separately, the number of RNA reads. We show that for both DNA-seq depth and RNA-seq depth, we are within the range of sequencing depth in which additional sequencing would add minimal new analysis windows in the dataset (Figure 1-figure supplement 4; lines 179-184; 715-722).

- 1. Correlation between replicates should be reported and displayed in a figure because low correlations might also point to too few reads. The authors mention: "This difference likely stems from lower variance between replicates in the present study, which increases power", but I couldn't find the data.*

We now report the correlations between RNA and DNA replicates within the current dataset and within the Lea et al., 2018 dataset (Figure 1-figure supplement 6). The between-replicate correlations in both our RNA libraries and DNA libraries are consistently high ($r \geq 0.89$).

- 1. The correlation between the previous and current K562 datasets is surprisingly low. Given that these datasets were generated in the same cell type, in the same lab, and using the same protocol, I expected a higher correlation, as seen in other massively parallel reporter assays. The fact that the correlations are almost identical for a comparison of the same cell and a comparison of very different cell types is also suspicious.*

Thanks for raising this point. We think it is in reference to our original Figure 1-Figure supplement 6, for which we now provide Pearson correlations in addition to R2 values (now Figure 1-Figure supplement 8). We note that this is not a correlation in raw data, but rather the correlation in estimated effect sizes from a statistical model for methylation-dependent activity. We now provide Pearson correlations for the raw data between replicates within each dataset (Figure 1-Figure supplement 6), which for the baseline dataset are all $r > 0.89$ for RNA replicates and $r > 0.98$ for DNA replicates, showing that replicate reproducibility in this study is on par with other published studies (e.g., Klein et al., 2020 report $r > 0.89$ for RNA replicates and $r > 0.91$ for DNA replicates).

We do not know of any comparable reports in other MPRA for effect size correlations between two separately constructed libraries, so it's unclear to us what the expectation should be. However, we note that all effect sizes are estimated with uncertainty, so it would

be surprising to us to observe a very high correlation for effect sizes in two experiments, with two independently constructed libraries (i.e., with different DNA fragments), run several years apart—especially given the importance of winner’s curse effects and other phenomena that affect point estimates of effect sizes. Nevertheless, we find that regions we identify as regulatory elements in this study are 74-fold more likely to have been identified as regulatory elements in Lea et al., 2018 ($p < 1 \times 10^{-300}$).

1. *The authors cite Johnson et al. 2018 to support their finding that merely 0.073% of the human genome shows activity (1.7% of 4.3%), but:*

a. the percent cited is incorrect: this study found that 27,498 out of 560 million regions (0.005%) were active, and not 0.165% as the authors report.

We have modified the text to clarify the numerator and denominator used for the 0.165% estimate from Johnson et al 2018 (lines 175-176). The numerator is their union set of all basepairs showing regulatory activity in unstimulated cells, which is 5,547,090 basepairs. The denominator is the total length of the hg38 human genome, which is 3,298,912,062 basepairs.

Notably, the denominator (the total human genome) is not 560 million—while Johnson et al (2018) tested 560 million unique ~400 basepair fragments, these fragments were overlapping, such that the 560 million fragments covered the human genome 59 times (i.e., 59x coverage).

b. other studies that used massively parallel reporter assays report substantially higher percentages, suggesting that the current study is possibly underpowered. Indeed, the previous mSTARR-seq found a substantially larger percentage of regions showing regulatory activity (8%). The current study should be compared against other studies (preferably those that did not filter for putatively active sequences, or at least to the random genomic sequences used in these studies).

We appreciate this point and have double checked comparisons to Johnson et al., 2018 and Lea et al., 2018. Our numbers are not unusual relative to Johnson et al., 2018 (0.165%), which surveyed the whole genome. Also, in comparing to the data from Lea et al., 2018, when processed in an identical manner (our criteria are more stringent here), our values of the percent of the tested genome showing significant regulatory activity are also similar: 0.108% in the Lea et al., 2018 dataset versus 0.082% in the baseline dataset. Finally, our rarefaction analyses (see our responses above) indicate that we are not underpowered based on sequencing depth for RNA or DNA samples. We also note that there are several differences in our analysis pipeline from other studies: we use more technical replicates than is typical (compare to 2-5 replicates in Arnold et al., 2013; Johnson et al., 2018; Muerdter et al., 2018), we measure DNA library composition based on DNA extracted from each replicate post-transfection (as opposed to basing it on the pre-transfection library: [Johnson et al., 2018], and we use linear mixed models to identify regulatory activity as opposed to binomial tests [Johnson et al., 2018; Arnold et al., 2013; Muerdter et al., 2018].

I find it confusing that the four sets of CpG positions used: EPIC, RRBS, NR3C1, and random control loci, add up together to 27.3M CpG positions. Do the 600 bp windows around each of these positions sufficient to result in whole-genome coverage? If so, a clear explanation of how this is achieved should be added.

Thanks for this comment. Although our sequencing data are enriched for reads that cover these targeted sites, the original capture to create the input library included some off target reads (as is typical of most capture experiments, which are rarely 100% efficient). We then sequenced at such high depth that we ultimately obtained sequencing coverage that encompassed nearly the whole genome. We now clarify in the main text that our protocol

assesses 27.3 million CpG sites by assessing 600 bp windows encompassing 93.5% of all genomic CpG sites (line 89), which includes off-target sites (line 149).

scatter plot showing the RNA to DNA ratios of the methylated (x-axis) vs unmethylated (y-axis) library would be informative. I expect to see a shift up from the x=y diagonal in the unmethylated values.

We have added a supplementary figure showing this information, which shows the expected shift upwards (Figure 1-figure supplement 9).

Another important figure missing is a histogram showing the ratios between the unmethylated and methylated libraries for all active windows, with the significantly differentially active windows marked.

We have added a supplementary figure showing this information (Figure 1-Supplementary Figure 10).

Perhaps I missed it, but what is the distribution of effect sizes (differential activity) following the various stimuli?

This information is provided in table form in Supplementary Files 3, 10, and 11, which we now reference in the Figure 2 legend (lines 365-366).

Minor changes

It is unclear what the lines connecting the two groups in Fig.3C represent, as these are two separate groups of regions.

We now clarify in the figure legend that values connected by a line are the same regions, not two different sets of regions. They show the correlation between DNA methylation and gene expression at mSTARR-seq-identified enhancers in individuals before and after IAV stimulation, separately for enhancers that are shared between conditions (left) versus those that are IFNA-specific (right). The two plots therefore do show two different sets of regions, which we have depicted to visualize the contrast in the effect of stimulation on the correlation on IFNA-specific enhancers versus shared enhancers. We have revised the figure legend to clarify these points (line 458-460).

L235-242 are unclear. Specifically - isn't the same filter mentioned in L241-242 applied to all regions?

Yes, the same filter for minimal RNA transcription was applied to all regions. We have modified the text (lines 264-265, 271, 275-277) to clarify that the enrichment analyses were performed twice, to test whether the target types were: 1) enriched in the dataset passing the RNA filter (i.e., the dataset showing plasmid-derived RNA reads in at least half the sham or methylated replicates; $n = 216,091$ windows) and 2) enriched in the set of windows showing significant regulatory activity (at $FDR < 1\%$; $n = 3,721$ windows).

To improve cohesiveness, the section about most CpG sites associated with early life adversity not showing regulatory activity in K562s can be moved to the supplementary in my opinion.

Thank you for this suggestion. Because ELA and the biological embedding hypothesis (via DNA methylation) were major motivations for our analysis (see Introduction lines 42-48; 75-79), and we also discuss these results in the Discussion (lines 518-520), we have respectfully

elected to retain this section in the main manuscript. We have added text in the Discussion explaining why we think experimental tests of methylation effects on regulation are relevant to the literature on early life adversity (lines 520-522), and have added discussion on limits to these analyses (lines 527-533).

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